



SAIR

Spatial AI & Robotics Lab

CSE 473/573

COMPUTER VISION & IMAGE PROCESSING

Chen Wang

Spatial AI & Robotics Lab

Department of Computer Science and Engineering

 **University at Buffalo** The State University of New York

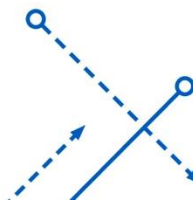
Introduction

University at Buffalo
Department of Computer Science and Engineering
CSE 473/573 - Computer Vision and Image Processing
Fall 2026

TuTh 3:30AM - 4:50AM

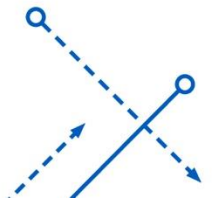
Location: Hoch 114

Aug 24, 2026 - Dec 7, 2026



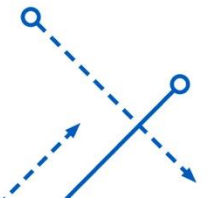
Instructor Information

Instructor:	Chen Wang, PhD. (https://sairlab.org)
Email:	Contacted through Piazza; All TA and instructor will receive msg



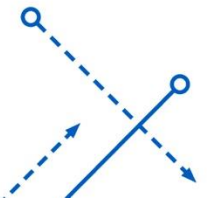
Piazza

- Piazza page
 - <https://piazza.com/buffalo/fall2026/cse473573a>
 - Access code: sairlab
- Slides will be put up regularly.
- Instructor and TAs will monitor Piazza questions



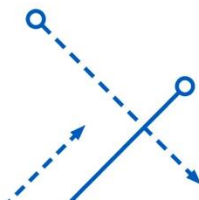
Course Information

- Lectures, Quizzes, Homework, Projects, and Mid and Final Exam.
- This course is an introduction to those areas of Artificial Intelligence that deal with fundamental issues and techniques of **computer vision** and **image processing**.
- The emphasis is on **physical, mathematical, and information-processing aspects** of the vision.
- Topics to be covered include **image formation, edge detection and segmentation, convolution, image enhancement techniques, extraction of features such as color, texture, and shape, object detection, 3-D vision, and their applications.**
- The material is based on graduate-level texts augmented with research papers.



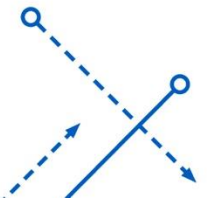
Overview

- The course will move faster than S26.
- **Self-discipline** is important.
- The emphasis of the course is to develop practical skills for solving **Computer Vision** and **Image Processing** problems
- **Fair evaluations: undergraduate and graduate students will be scored separately**
- Academic Integrity (AI) will be taken seriously and zero tolerance to any cheating
 - Please work on projects independently.
- ChatGPT and Copilot are not allowed in quizzes, projects, exams.



Prerequisites:

- CSE 203
 - Data Structure
- Strong Knowledge of Linear Algebra
- Strong Programming Experience in Python
- Git & GitHub:
 - <https://www.coursera.org/learn/introduction-git-github>



Grading

<u>Weighting</u>	<u>Assessment</u>	<u>Number</u>
20%	Midterm Exam	2
20%	Quizzes	5
30%	Projects	3
30%	Final Exam: Hoch 114 12/15/2026, Tu 3:30PM - 5:30PM	1

Today: Quiz 0

Grading

- All assignments will be graded out of 100 points

1. Undergraduate Version

Score	Letter Grade	Score	Letter Grade
[85,100]	A	[52,58)	C+
[80,85)	A-	[46,52)	C
[73,80)	B+	[41,46)	C-
[65,73)	B	[40,41)	D
[58,65)	B-	[0,40)	F

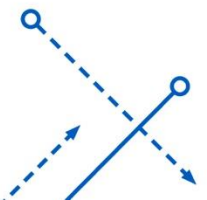
2. Graduate Version



Score	Letter Grade	Score	Letter Grade
[87,100]	A	[55,60)	C+
[82,87)	A-	[50,55)	C
[75,82)	B+	[46,50)	C-
[67,75)	B	[45,46)	D
[60,67)	B-	[0,45)	F

Quizzes

- Class attendance and participation is expected.
 - Random Quizzes.
- Quizzes will be given
 - **Class time only; online through UB Learns.**
- **You'll need a laptop.**
 - **Require Respondus LockDown Browser + Webcam**
- **Failure Tolerance:** We will drop **ONE Quiz** Grade, whichever results in a **higher** overall grade.
- **No makeup quizzes (no exceptional cases).**
- You may use the “failure tolerance” policy.

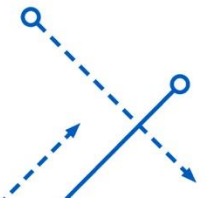


Projects

- All projects will be submitted via
 - **UB AutoLab**: Used for grading and ranking.
 - MOSS (Measure of Software Similarity) is integrated directly into UB Autolab for plagiarism detection.
- Our Observation
 - Students get higher scores since AI coding is available
 - Knowledge test worse: Q&A during presentation
- Changes:
 - Add hidden test cases.
 - Evaluate accuracy, efficiency, and memory usage; instead of whether task is completed.
 - Final scores will be based on performance ranking.

Late Submission Policy

- Completed project deliverables are to be submitted by their deadline (11:59pm).
- Grace days: You will be allowed a total of **3 grace days** for projects throughout the semester
- Submissions beyond the 3 grace days will reduce your grade by 50%;
- No individual project will be accepted after 3 days late. No additional late days allowed.



Exam Policy

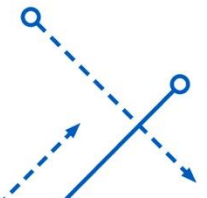
- No makeup exams will be given except in **extreme circumstances** and when consistent with University Policy.
- Notify your instructor & TA **1 month prior to the exam** via **piazza** if you are going to miss it. If it is medically impossible for you to give prior notice, please obtain a **note from a physician detailing the period** (with reasons) you were medically incapable of communicating with the instructor.
- You are responsible for knowing about the **exam date**. Please plan your travel and other activities accordingly.

Regrading for Errors

- Regrade requests should be submitted to Piazza, and are due **no later than 2 days** after the scores are posted.
- When work is submitted for regrade, the entire work may be regraded, **which may result in a lower grade.**
- Work done in pencil may not be considered for regrading.

Option B

- Received 10+ absence notice for Fall 2026.
- Feedback about no compulsory attendance.
 - Release videos regularly
 - Turn lectures to office hours.
 - Only require class time on **5 quiz days**.
 - Teaching the most important topics in final exam.
 - The others are the same.
- We will create a **voting** on Piazza for this.
 - We will go for Option B, if
 - More than **80% students** agree.



Disabilities

- If you have a diagnosed disability (physical, learning, or psychological) that will make it difficult for you to carry out the course work as outlined, or that requires accommodations such as recruiting note-takers, readers, or extended time on exams or assignments, please **advise the instructor during the first two weeks of the course** so that we may review possible arrangements for reasonable accommodations.
- In addition, if you have not yet done so, contact the **Office of Disability Services**.

Academic Integrity

- All work must be your own
 - Do not take the answers, words, ideas or research findings of other people as yours; cite and acknowledge properly, and develop your own ideas.
 - No cheating
 - According to departmental policy, any violation of academic integrity will result in an “F” for the course, and termination of departmental financial scholarship.
 - Tools will be used to check similarity. **Similar submissions will result in “F” for all involved parties.**
- Use of code from online, e.g., GitHub, must include a proper and clearly visible attribution in your report.

How to Fail the Course

- Don't meet the prerequisites
- Don't pay attention to information on Piazza
- Start your project at the last minute
- Wait until the deadline to submit for the first time
- Don't read the syllabus carefully (Academic Integrity violation and late submission policy etc.)
- **Cheat (please check AI policy of the university and department if you are new)**
- <https://www.buffalo.edu/academic-integrity/policies.html>

Questions?

- **Instructors will stay for a few minutes after each lecture. Simply ask!**
- Syllabus and Slides are on Piazza.



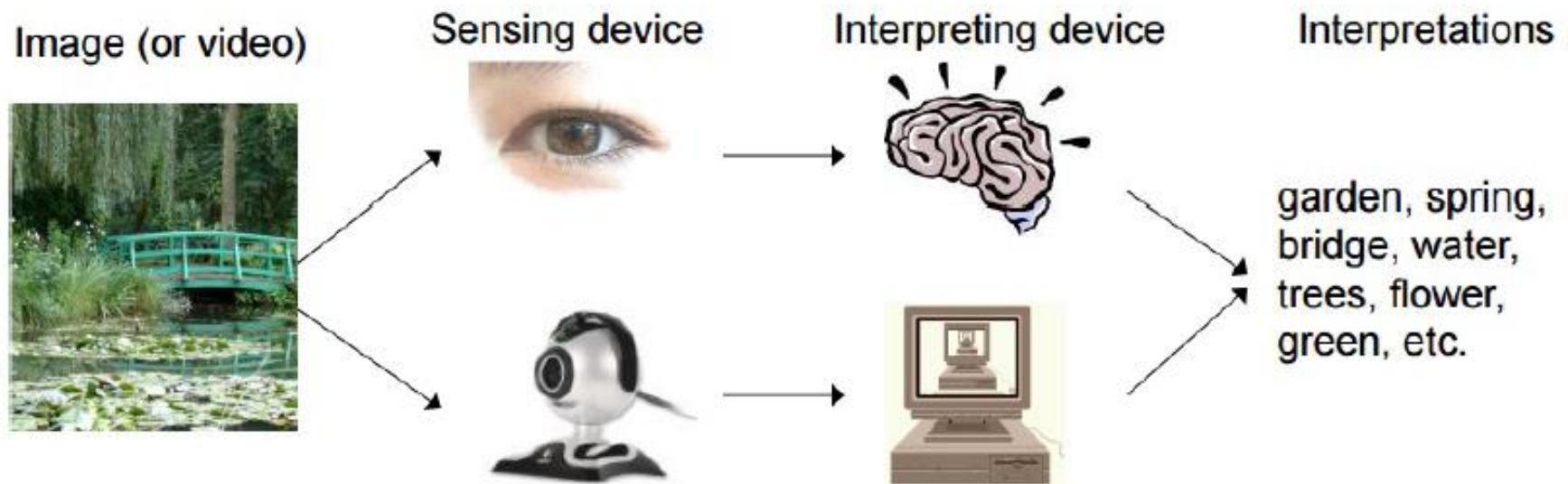
Content

- Definition
- Research Topics
- Applications
- Quiz 0
 - UB Learn, Academic Integrity Letter
 - To help you familiar with the Quiz system.
 - Future installation issues will not be given chances.

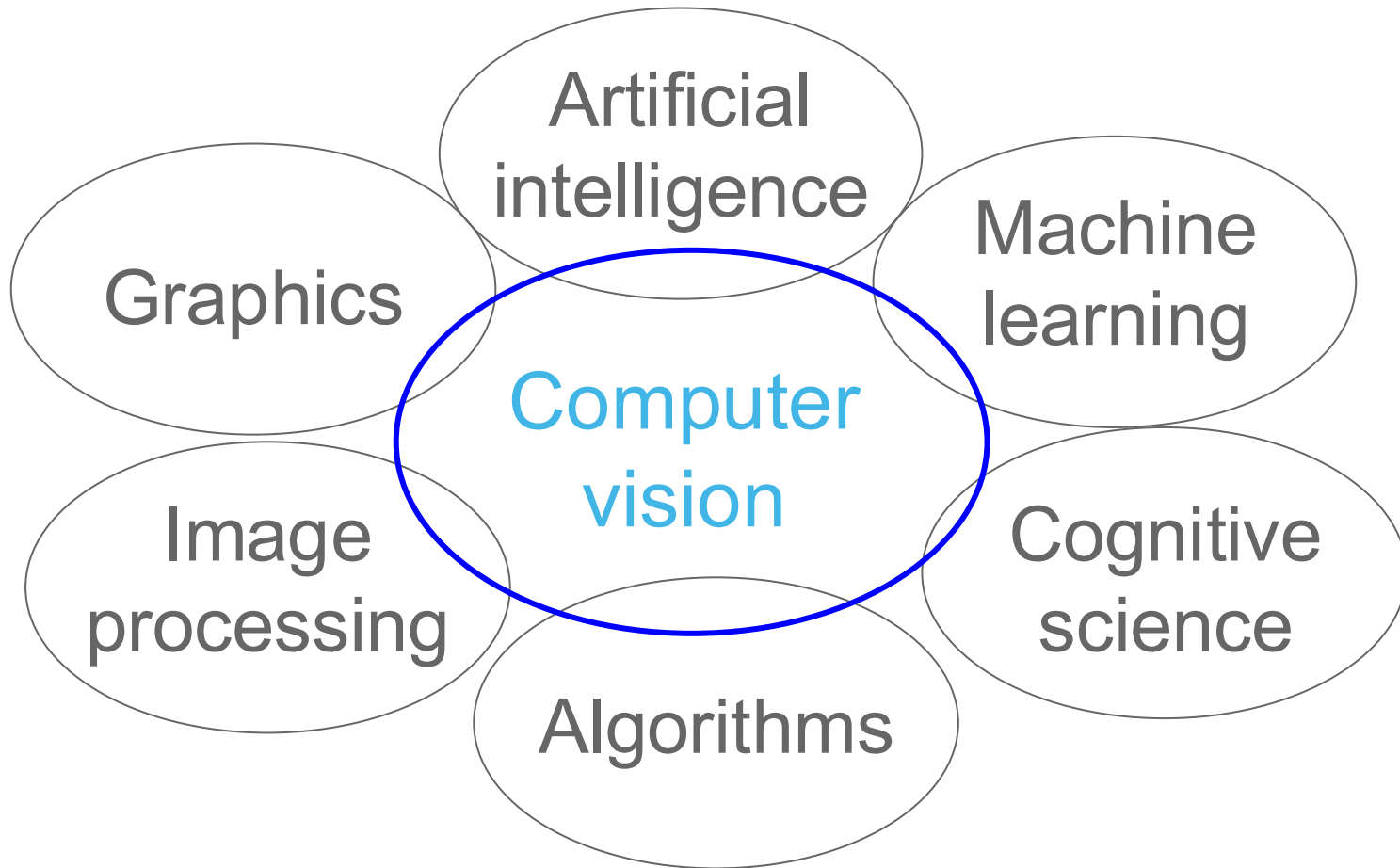
What is Computer Vision

- **Computer vision** is a field of **computer science**
 - works on enabling **computers** to see,
 - identify and process images in the same way that human **vision** does, and
 - then provide proper output.
- It is like imparting human intelligence and instincts about vision to a **computer**.

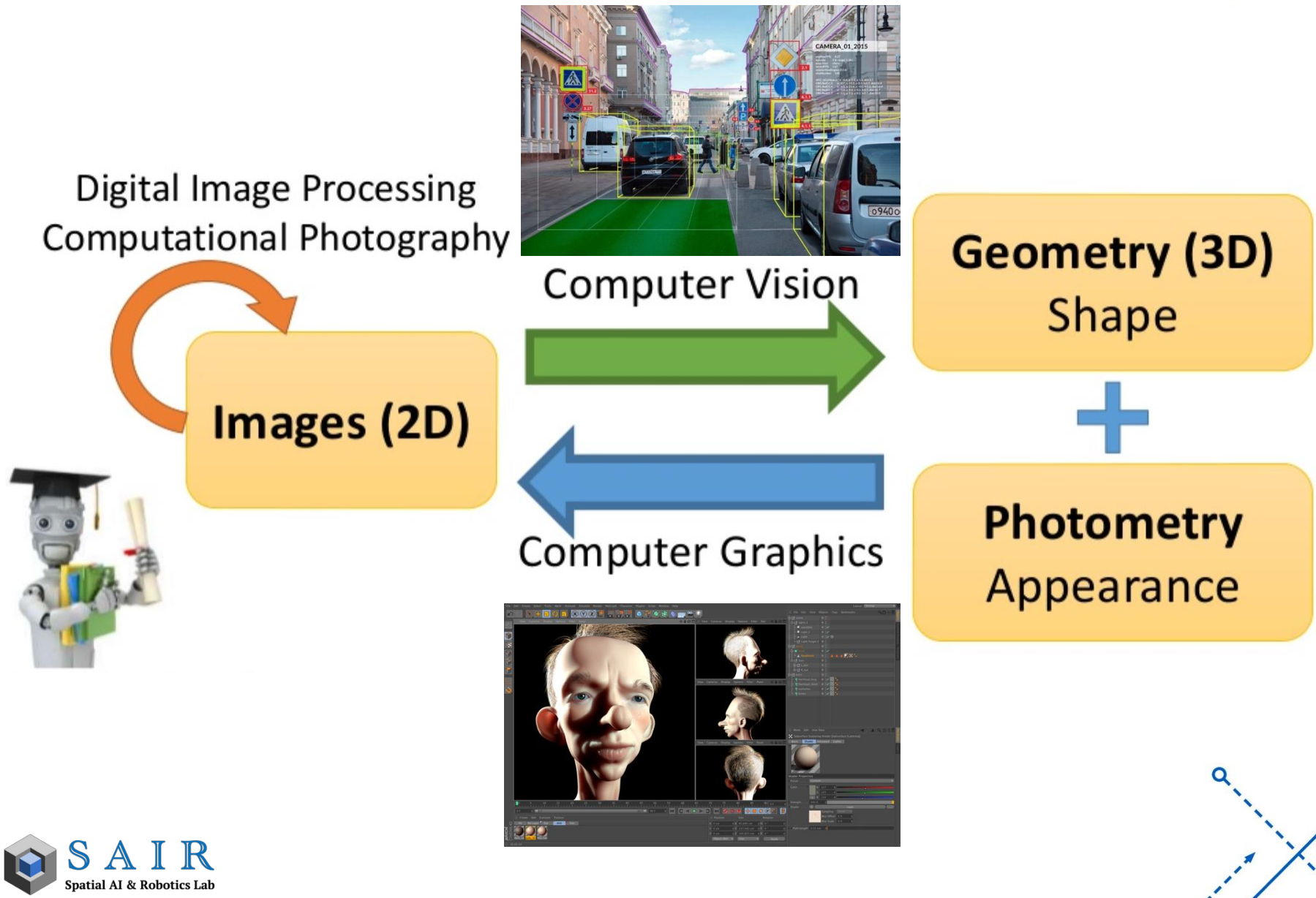
Computer Vision vs Human Vision



Related disciplines



Computer Vision vs Computer Graphics



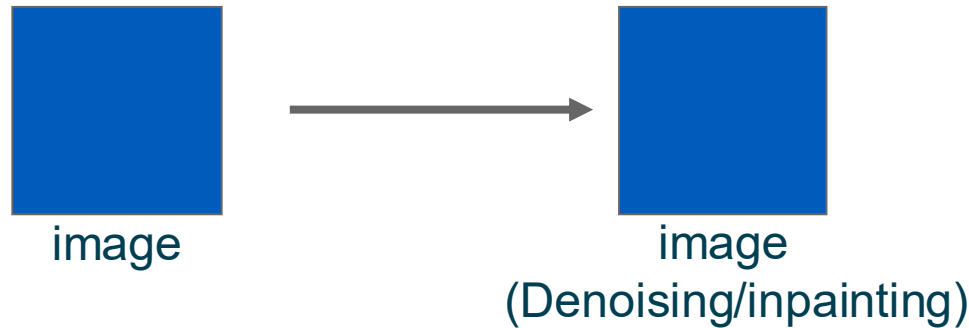
Example: Visual Effects need CV + CG



Image Processing vs. Computer Vision

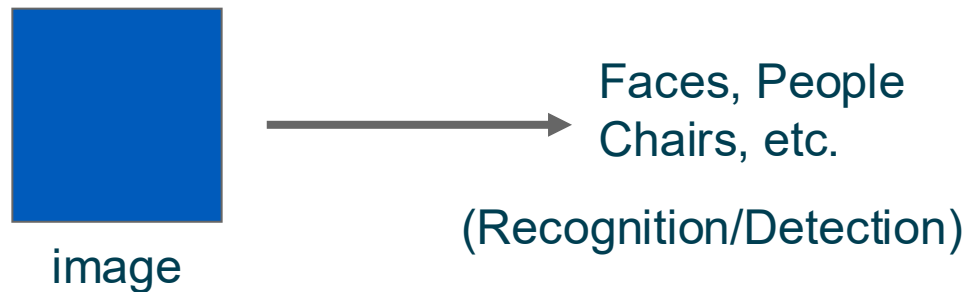
- Image Processing

- Research area within electrical engineering/signal processing
- Focus on syntax, low-level features



- Computer Vision

- Research area within computer science/artificial intelligence
- Focus on semantics, symbolic or geometric descriptions



Why learn Computer Vision?

- One third of human brain devoted to vision.
- 80% information is from vision.

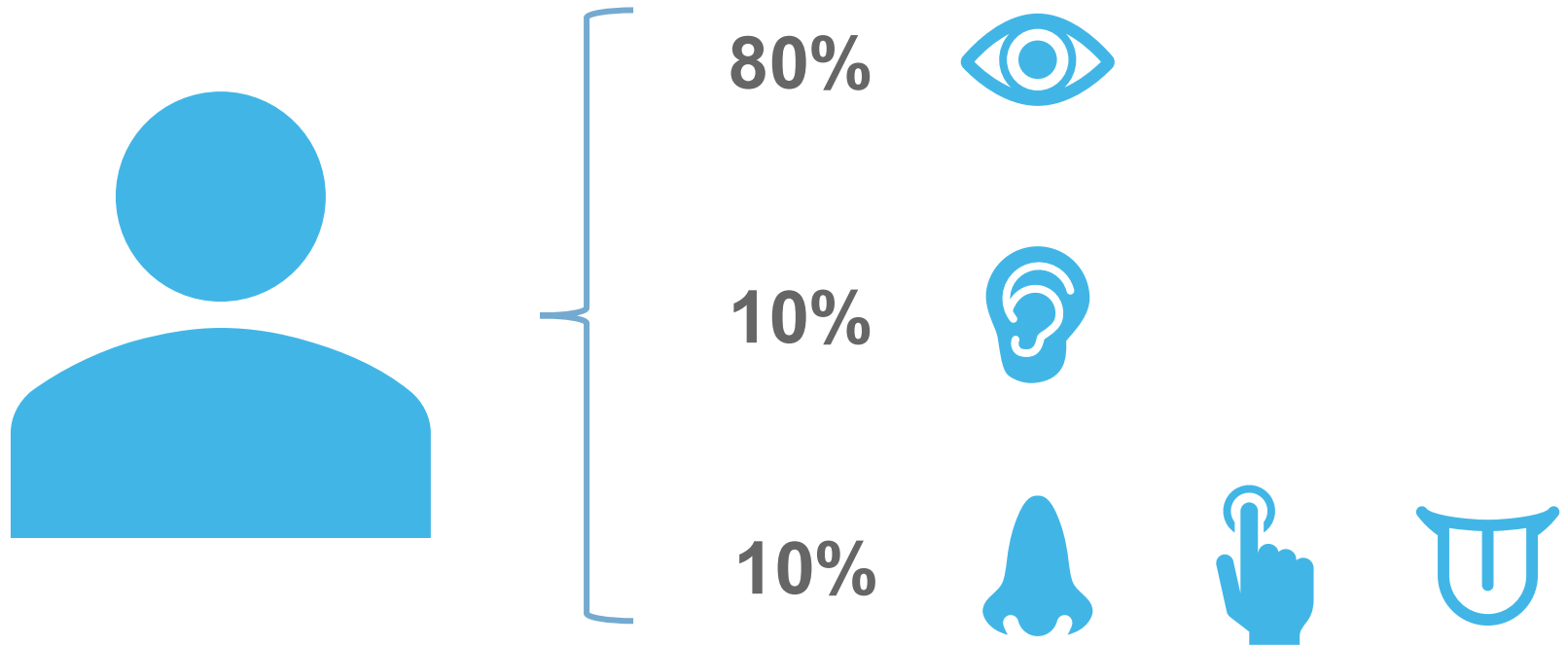


Image & Goal

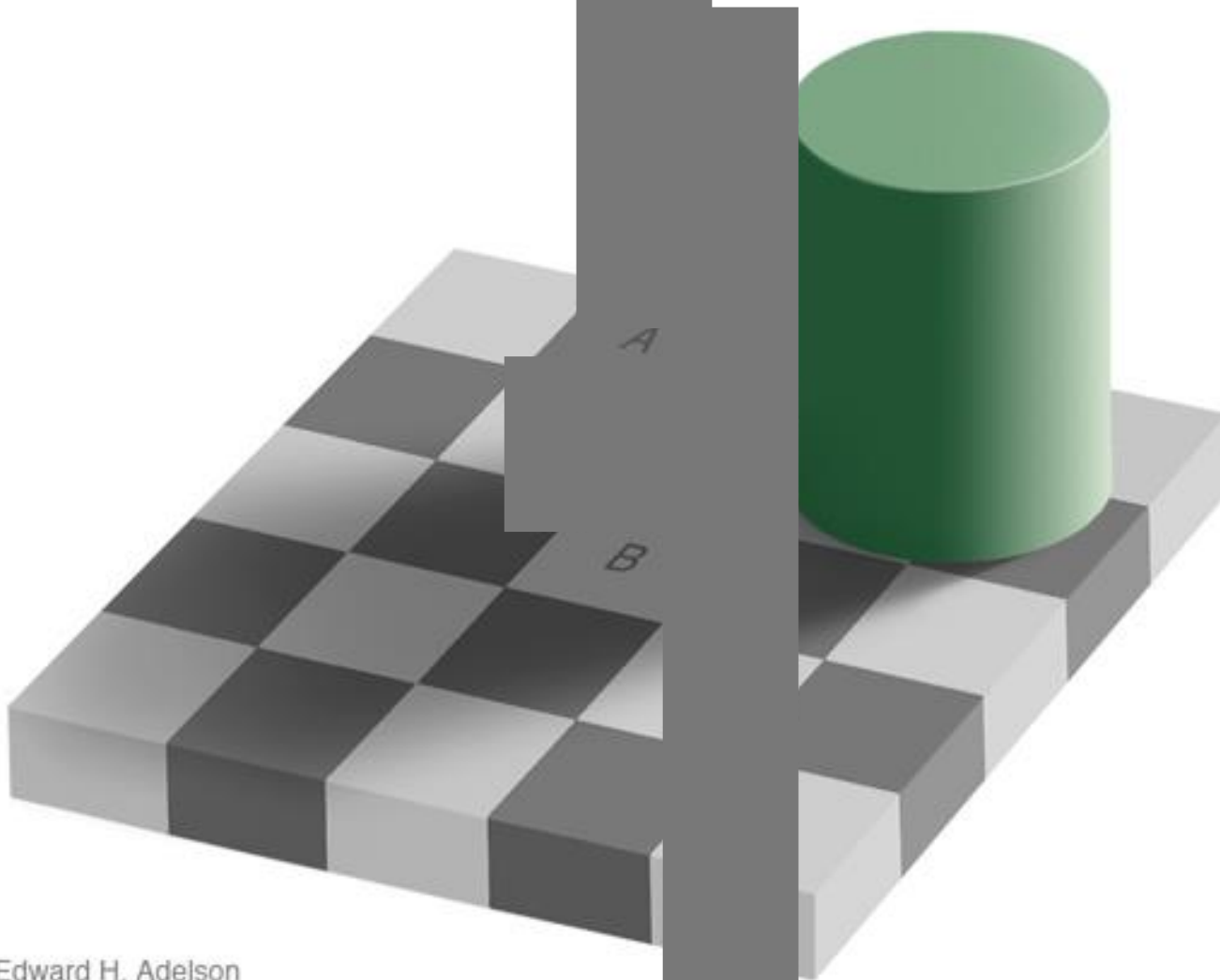


An image is an array of numbers (pixels).

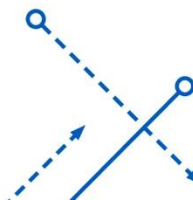
0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

- To Bridge the Gap between Pixels and Meaning

Which one is brighter? A or B?



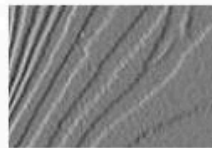
Edward H. Adelson



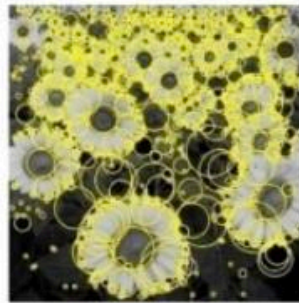
Feature Extraction



- Basic image processing and image formation



Filtering, edge detection



Feature extraction

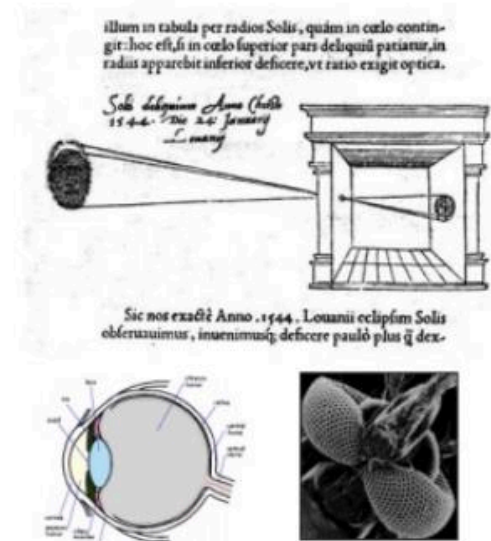
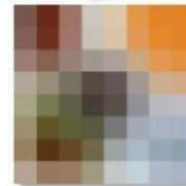
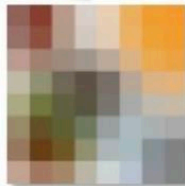
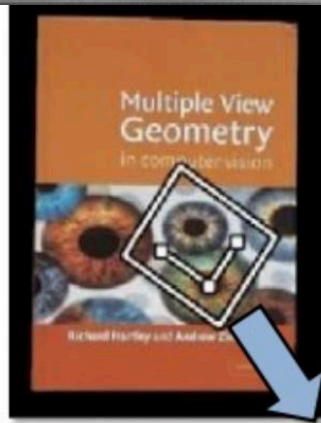


Image formation

Feature Matching



Recognition and Detection

Classification



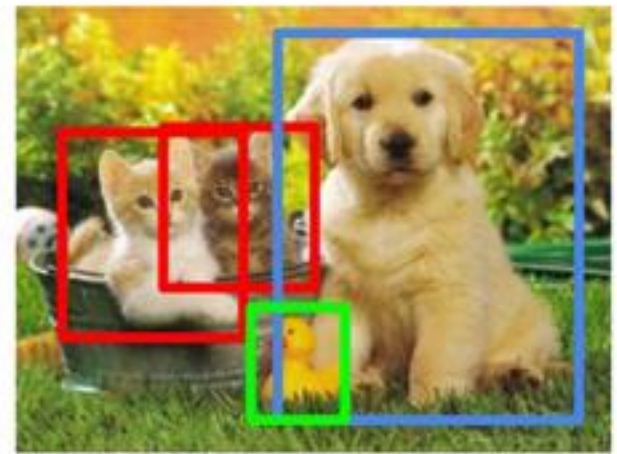
CAT

**Classification
+ Localization**



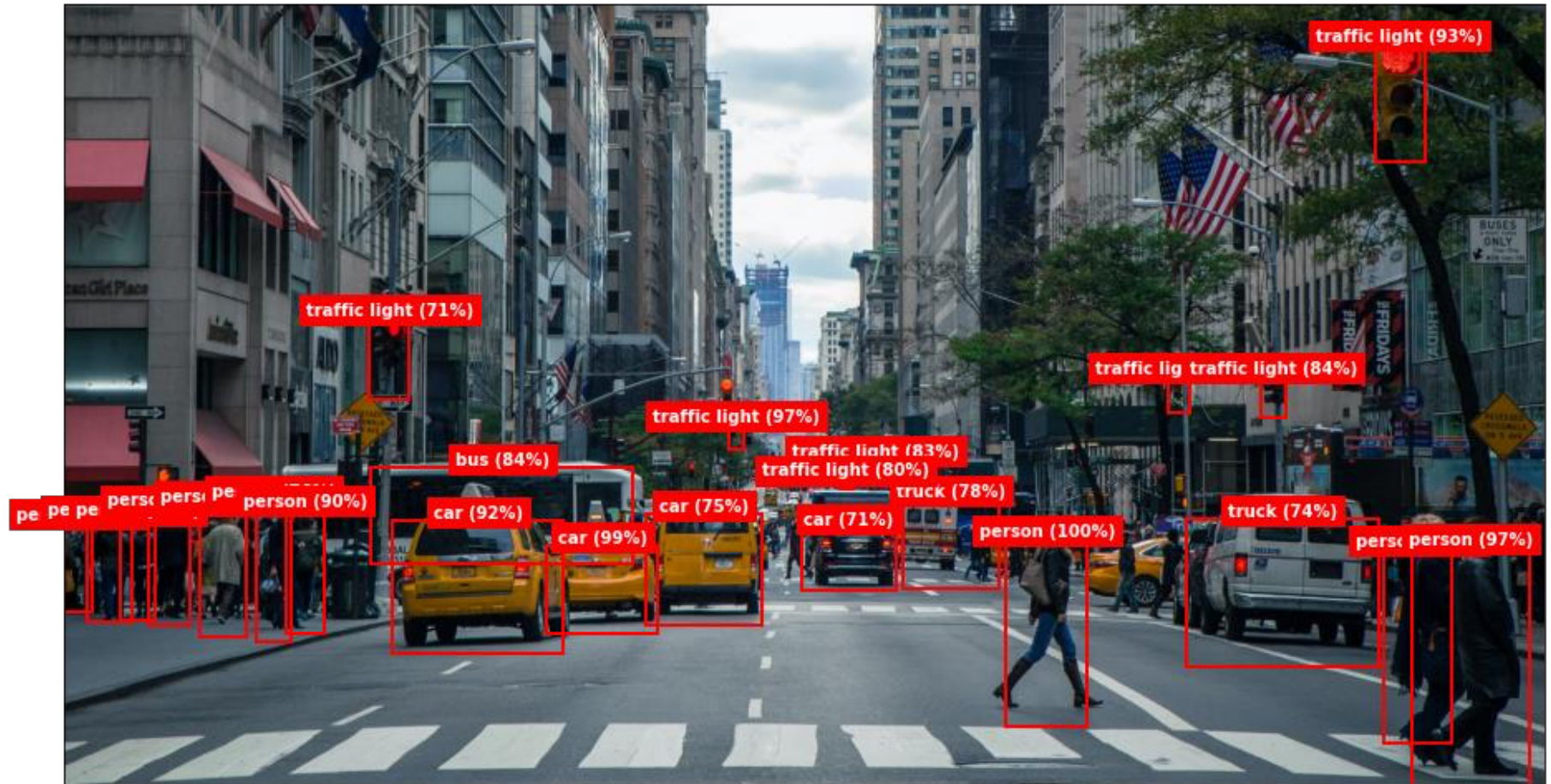
CAT

Object Detection



CAT, DOG, DUCK

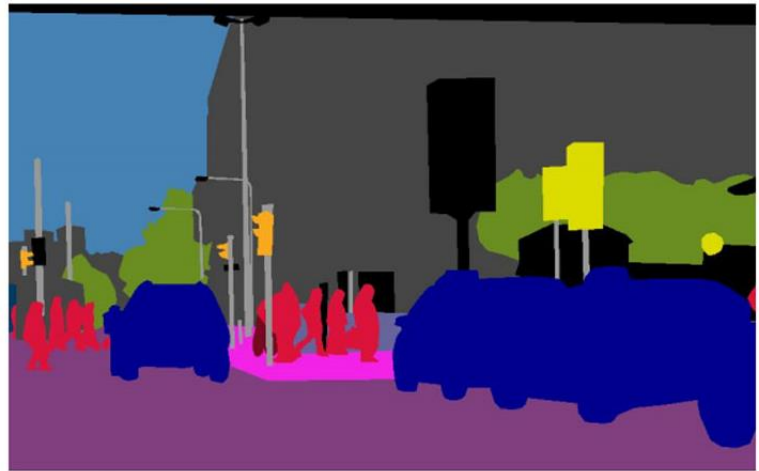
Detection



Segmentation



(a) image



(b) semantic segmentation



(c) instance segmentation

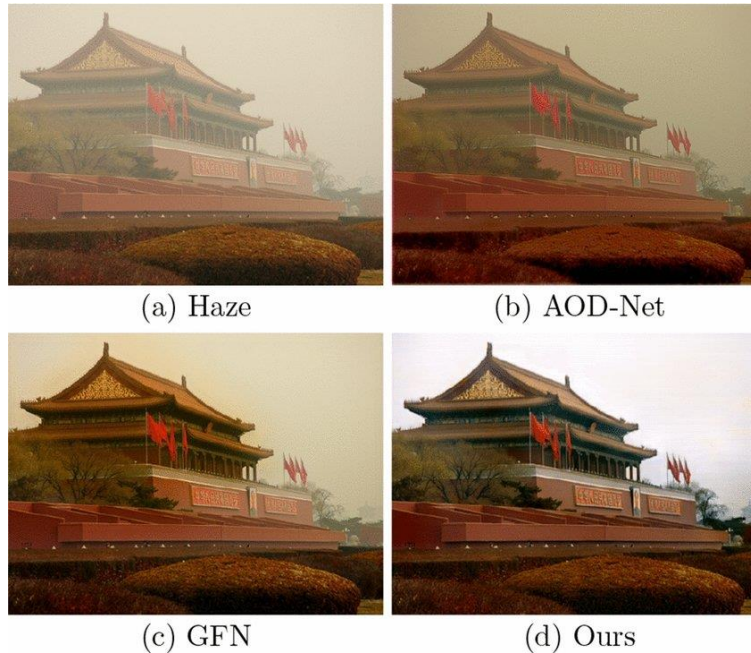
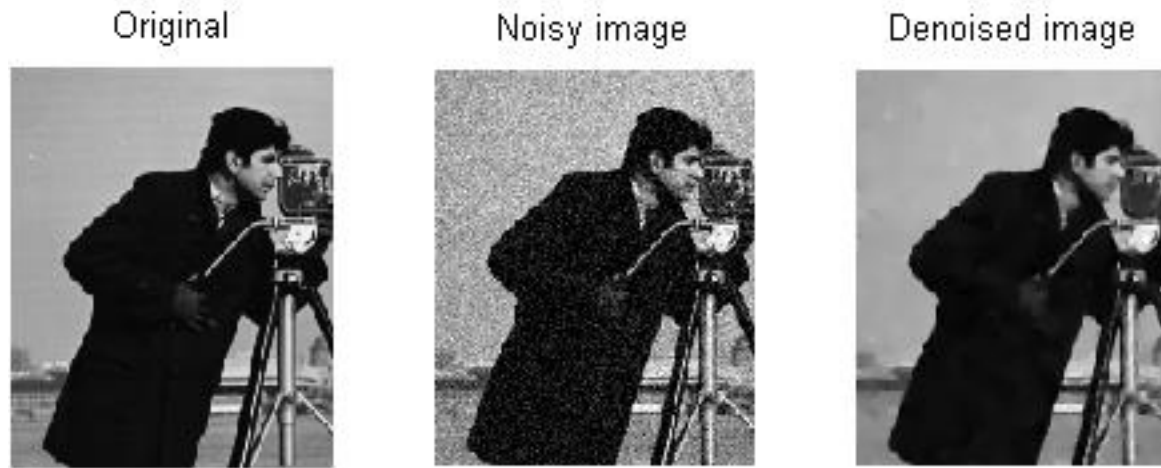


(d) panoptic segmentation

Inpainting



Denoising (Dehazing, Deblurring)



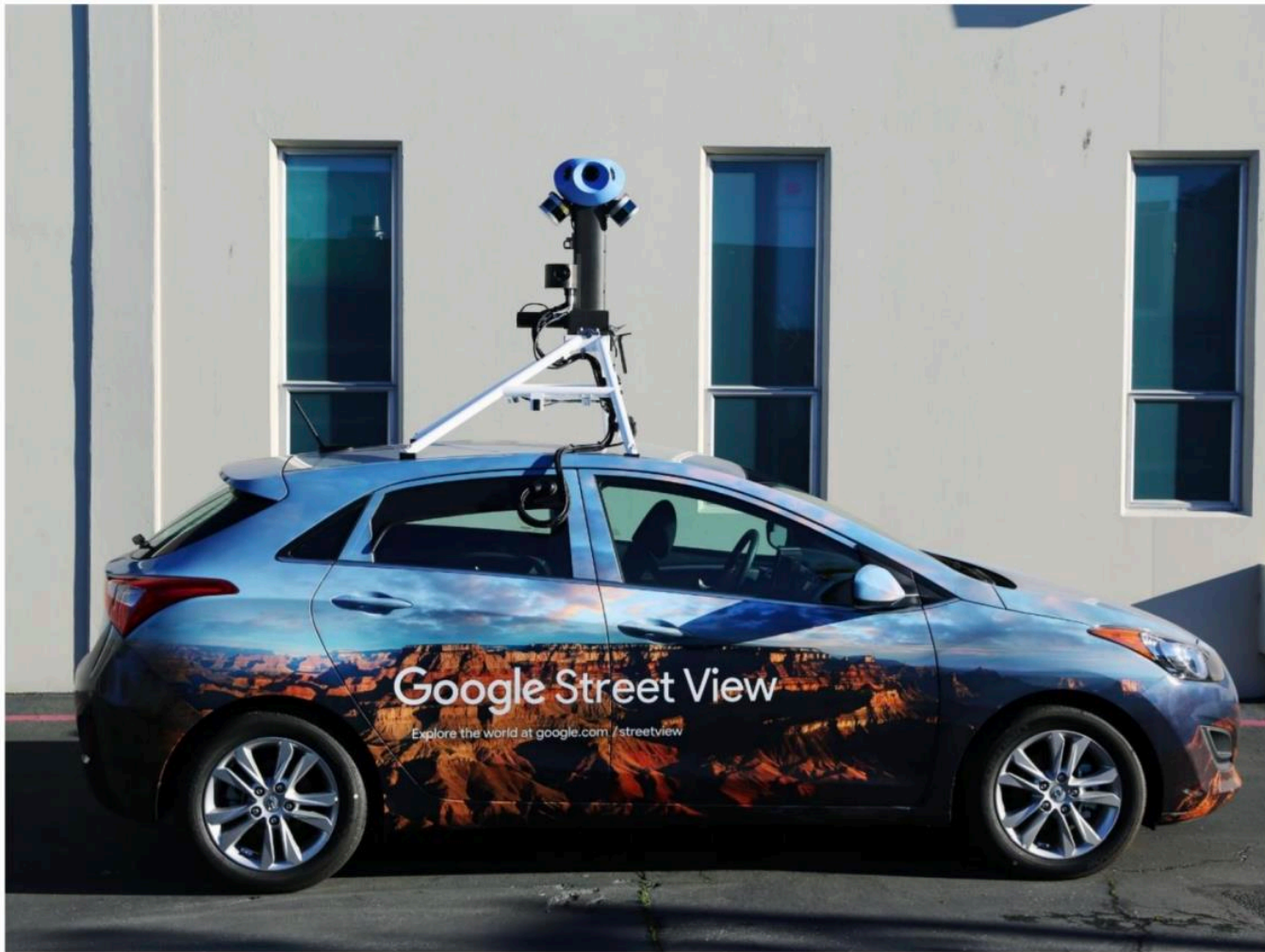
Super Resolution



Image Stitching: Panorama



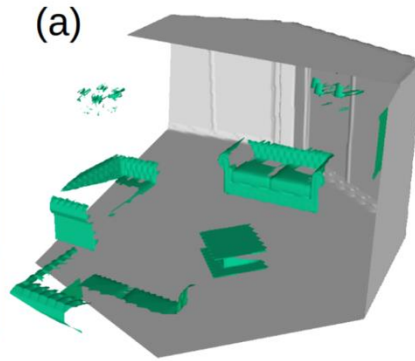
Example: Google Street View



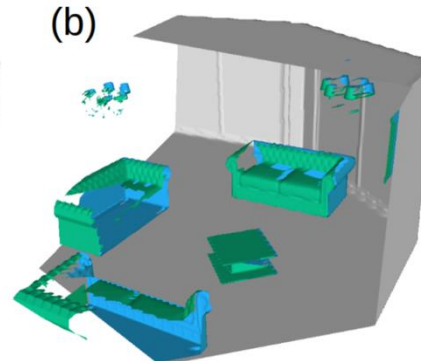
3D Reconstruction (RGB/RGBD)



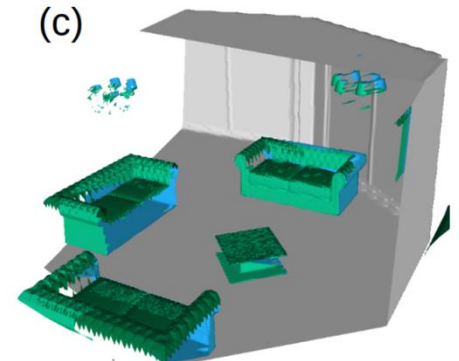
RGB Image



2.5D Object Surfaces



Multi-layer Surfaces



Multi-layer and
Virtual-view Surfaces

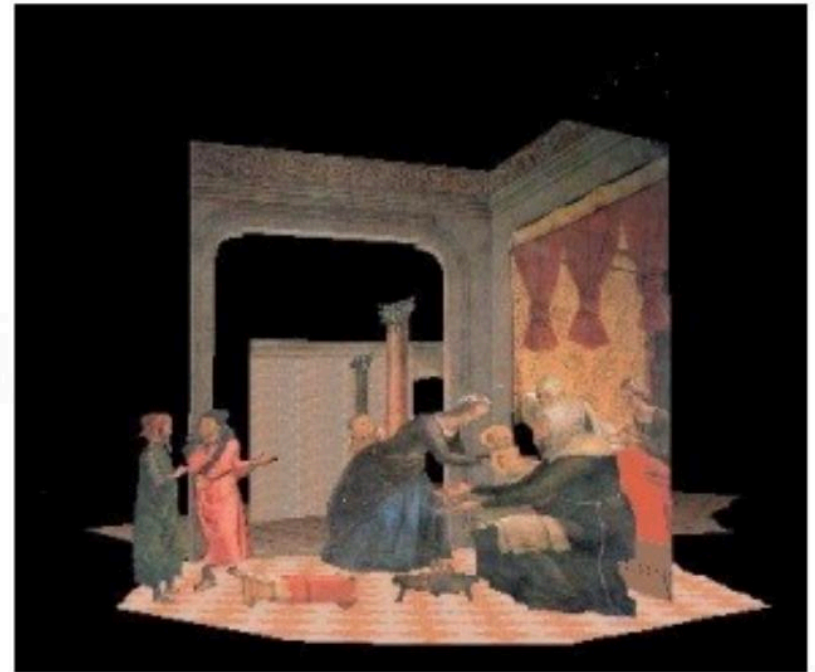
32 Views



0.51 s



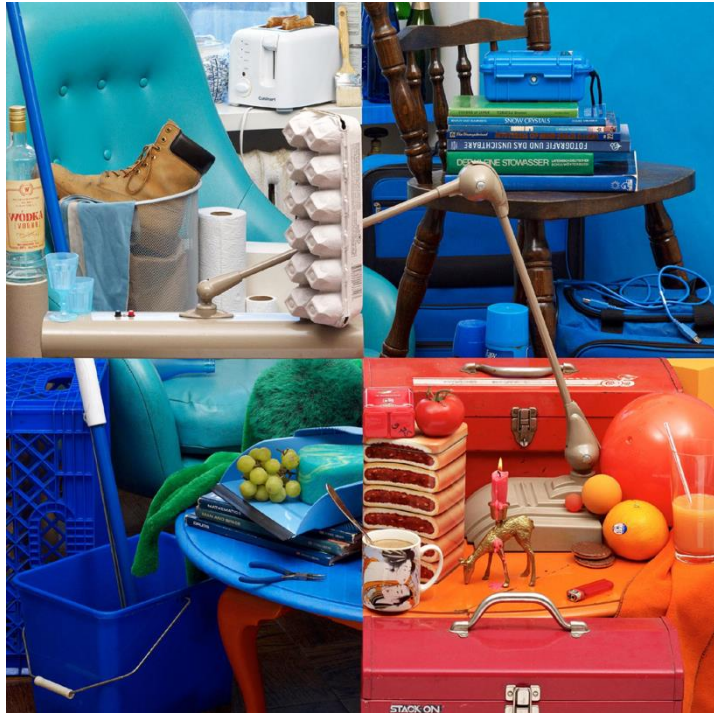
Single View Modeling



Can you do single view modeling?



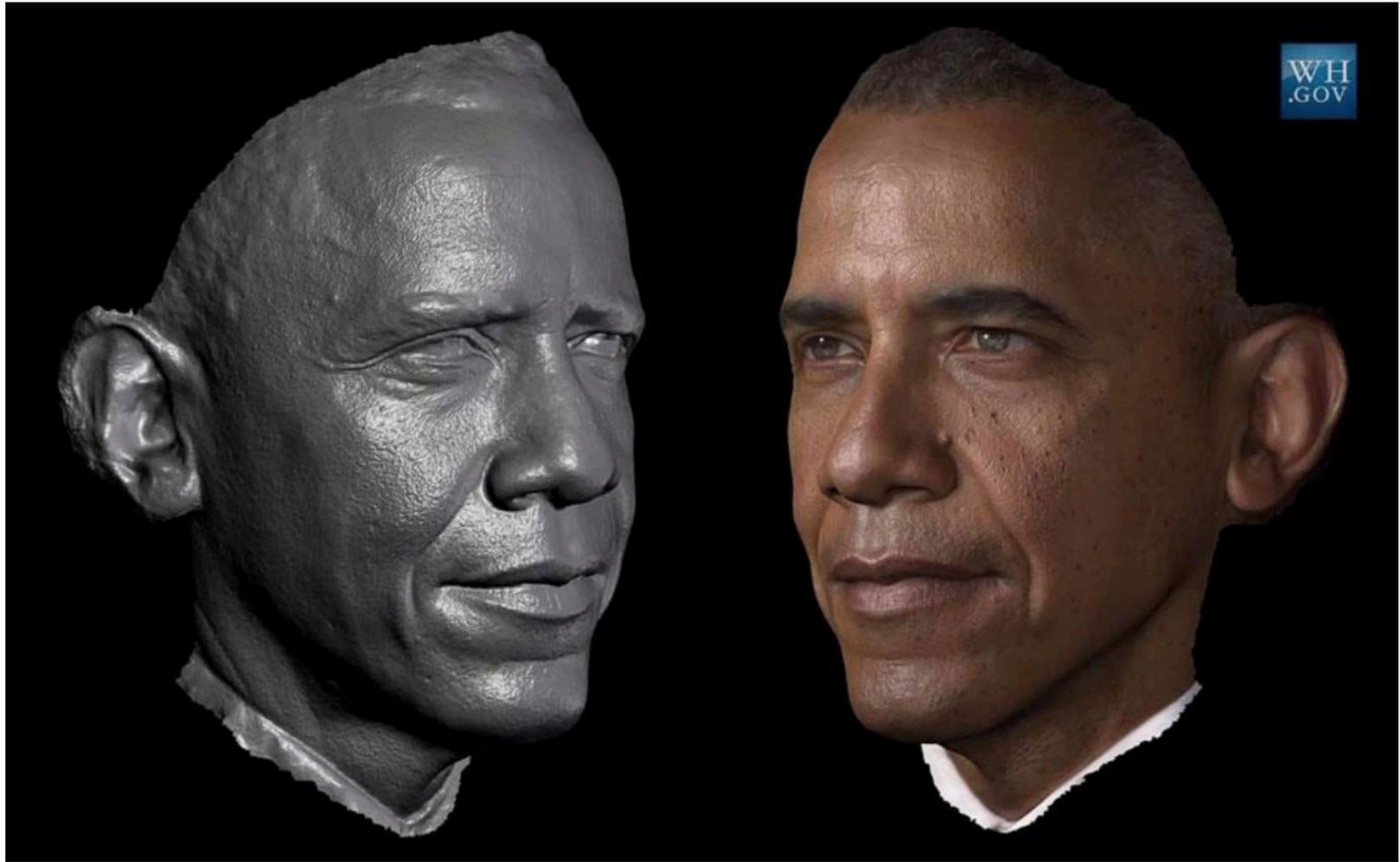
Can you do single view modeling?



Multi-Camera

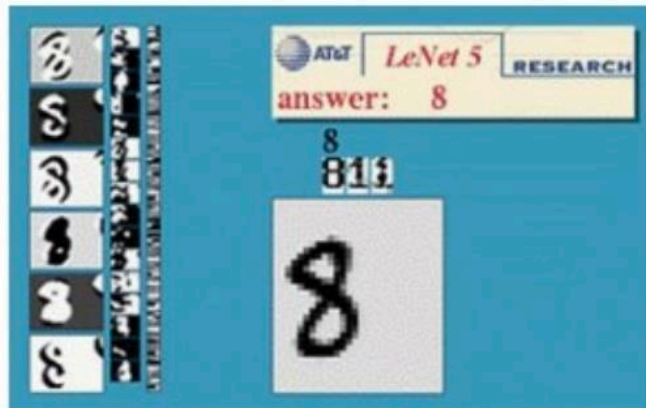


3D Scanning



Everyday Applications - OCR

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs

<http://www.research.att.com/~yann/>

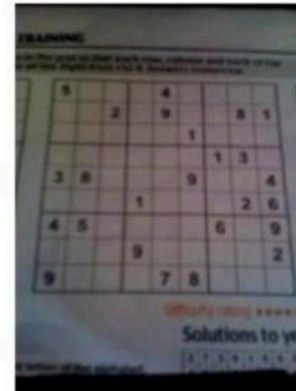


License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



Automatic check processing



Sudoku grabber

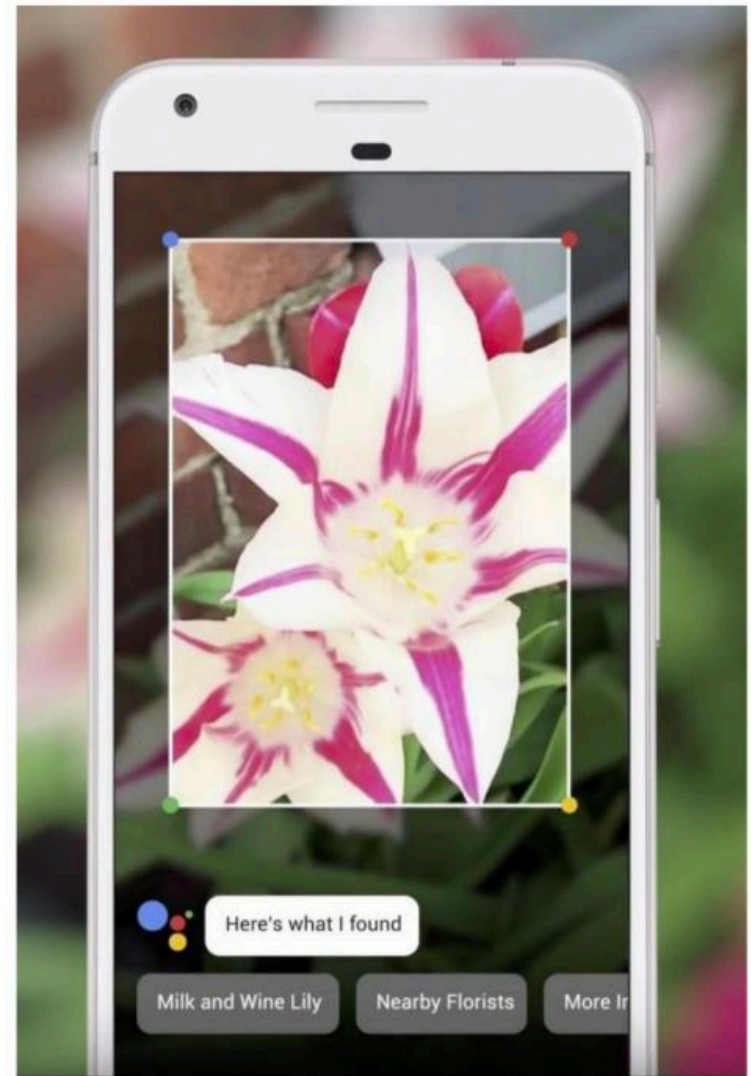
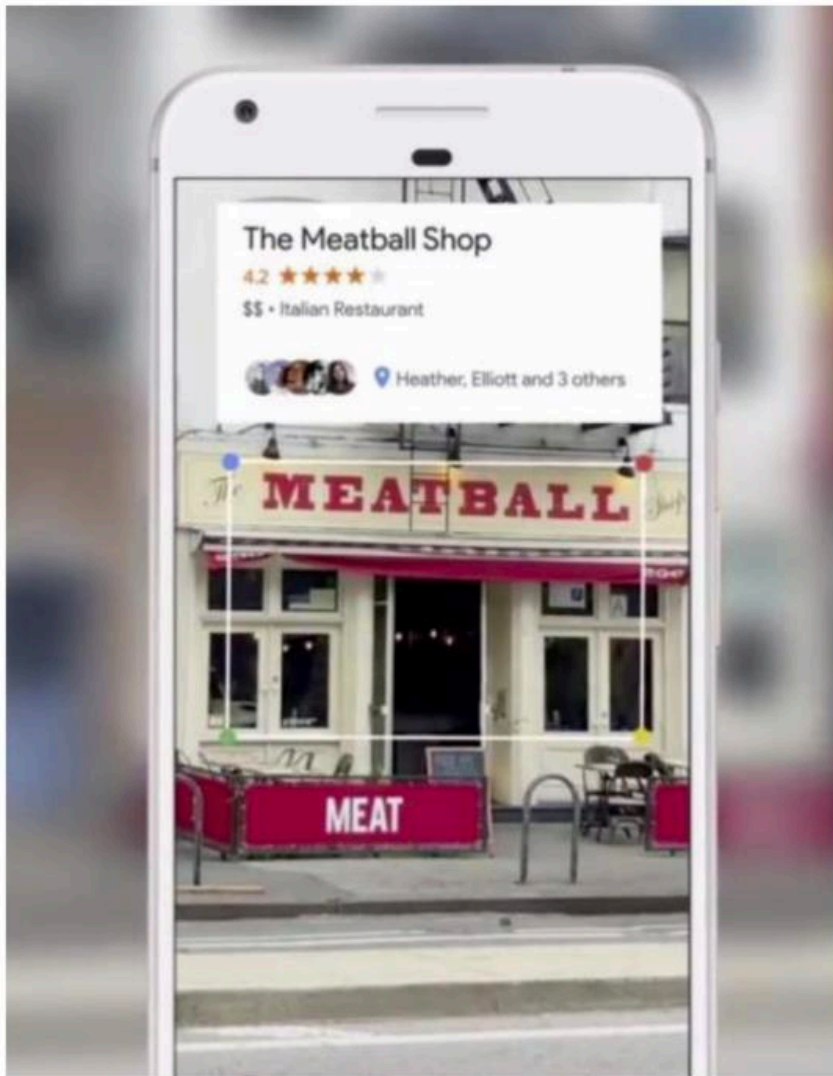
<http://sudokugrab.blogspot.com/>

Source: S. Seitz

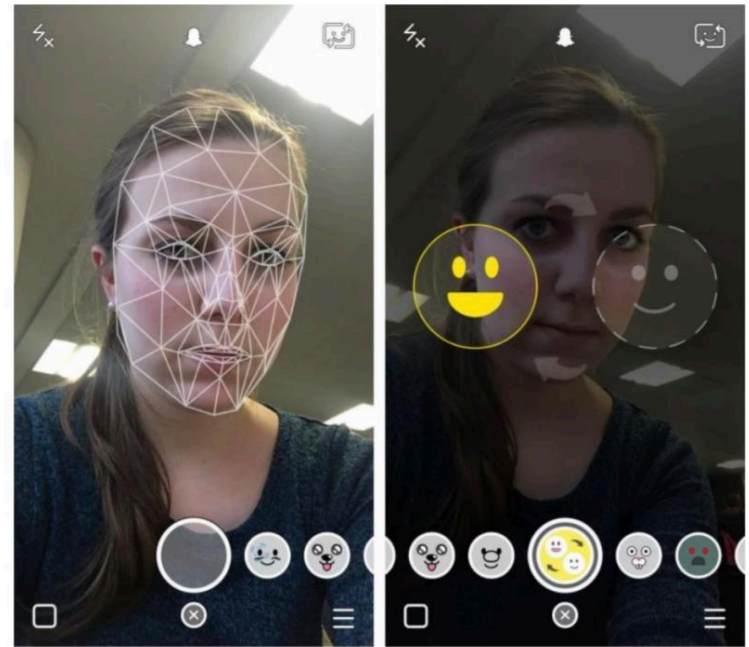
Object Recognition



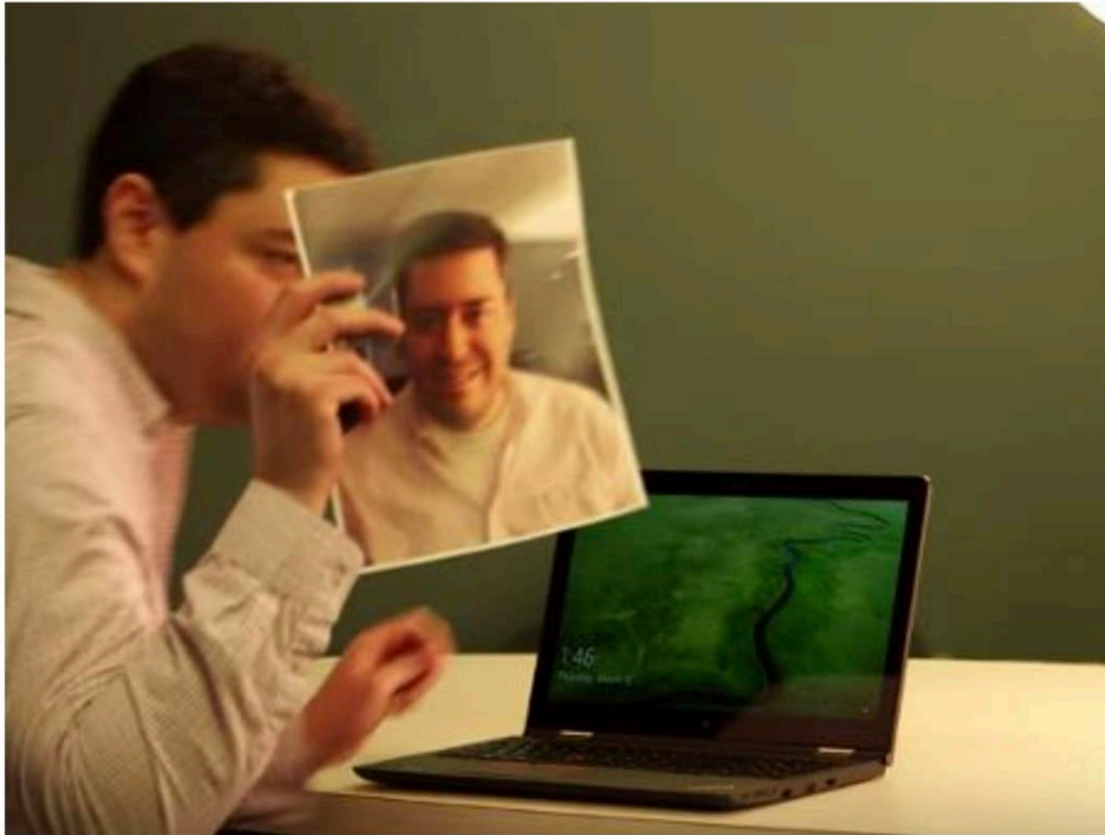
Visual Search: Google Lens



Face Detection



Face Recognition



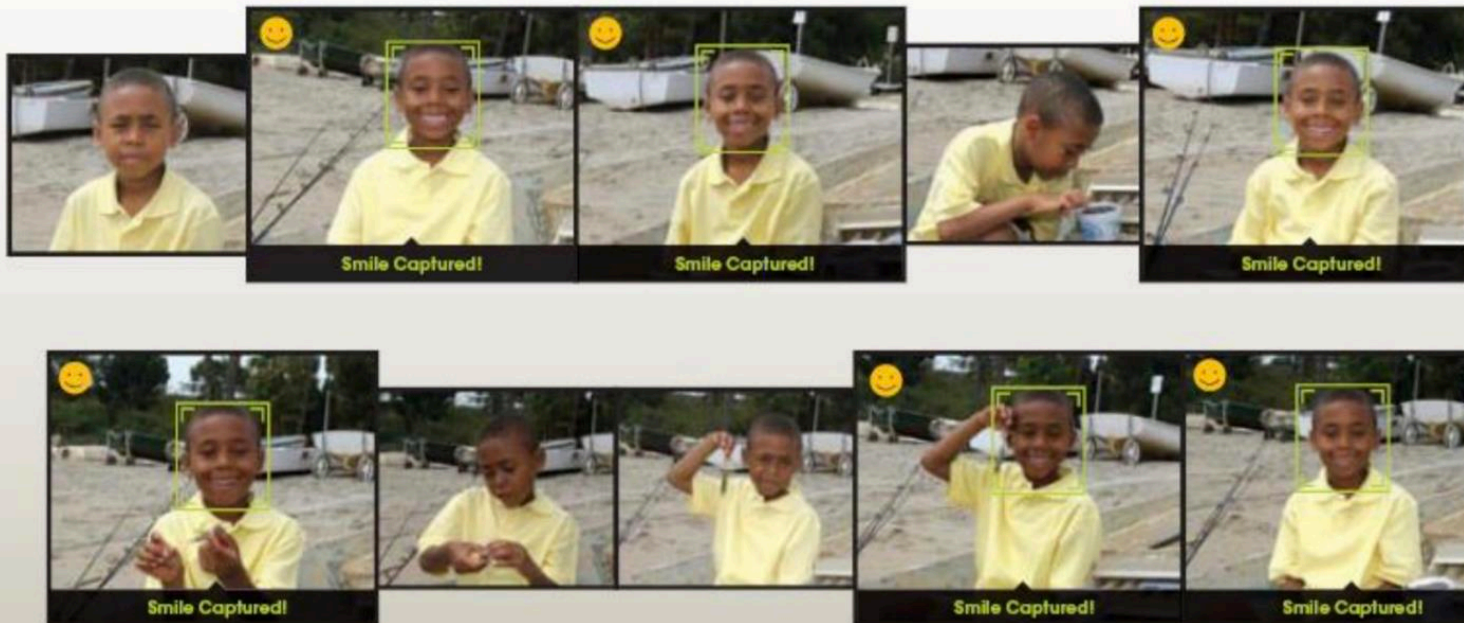
How to solve this problem?



Smile Detection

The Smile Shutter flow

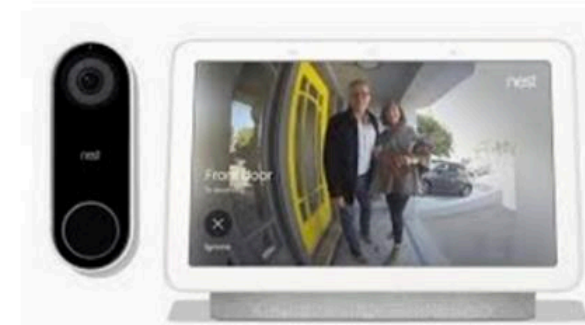
Imagine a camera smart enough to catch every smile! In Smile Shutter Mode, your Cyber-shot® camera can automatically trip the shutter at just the right instant to catch the perfect expression.



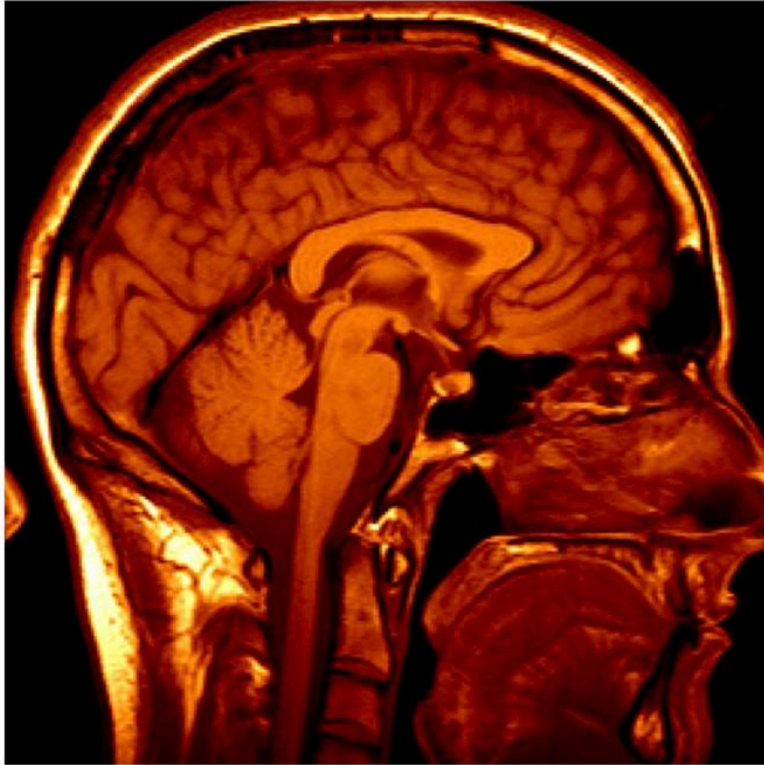
Entertainment



Video Surveillance



Medical Imaging

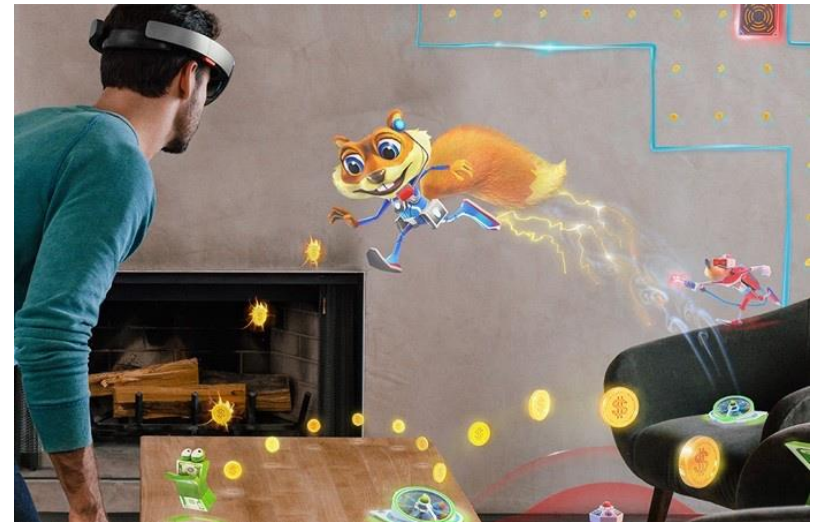
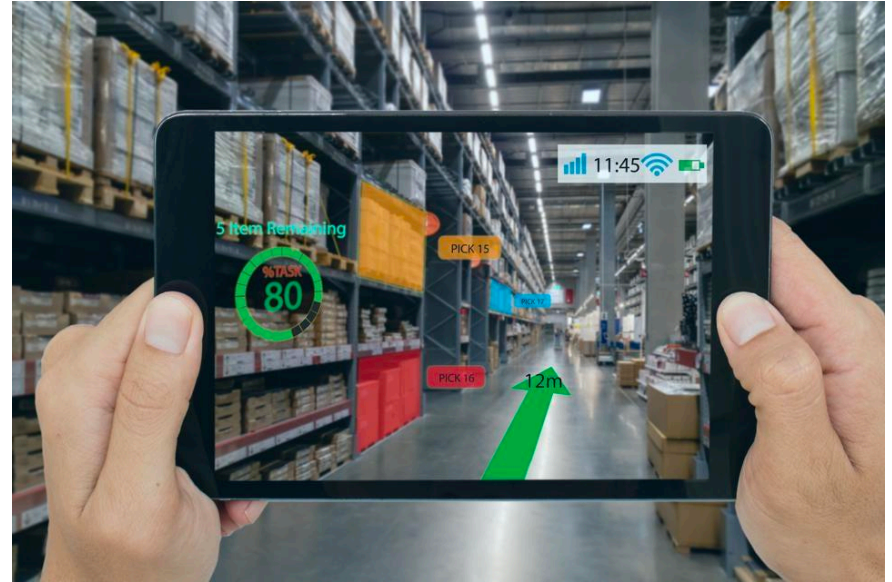


3D imaging
MRI, CT

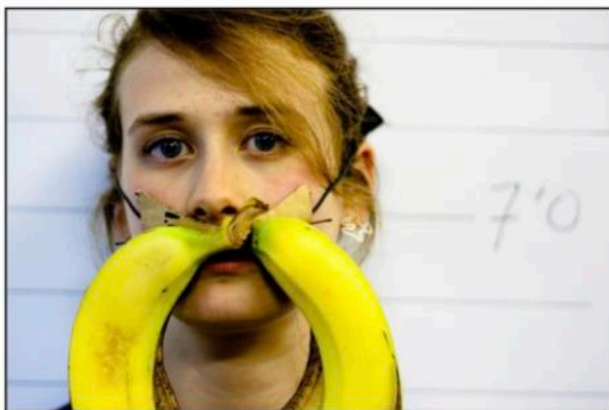


Image guided surgery
[Grimson et al., MIT](#)

Virtual/Augmented/Mixed Reality



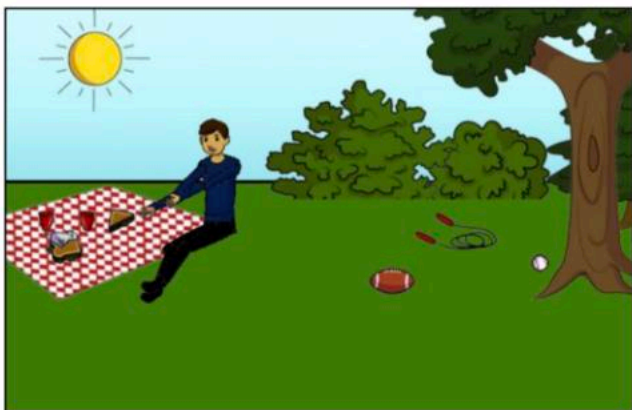
Visual Question and Answering (VQA)



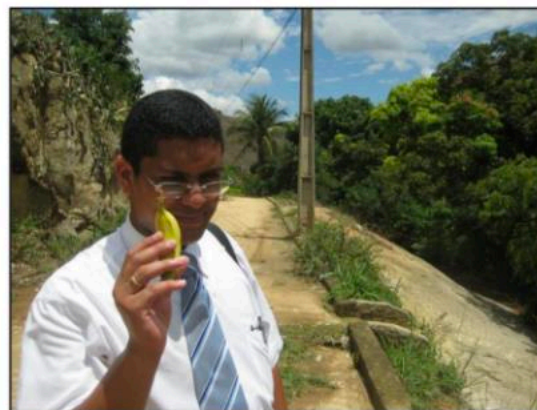
What color are her eyes?
What is the mustache made of?



How many slices of pizza are there?
Is this a vegetarian pizza?



Is this person expecting company?
What is just under the tree?



Does it appear to be rainy?
Does this person have 20/20 vision?

Autonomous Vehicle Navigation



Vision in Space

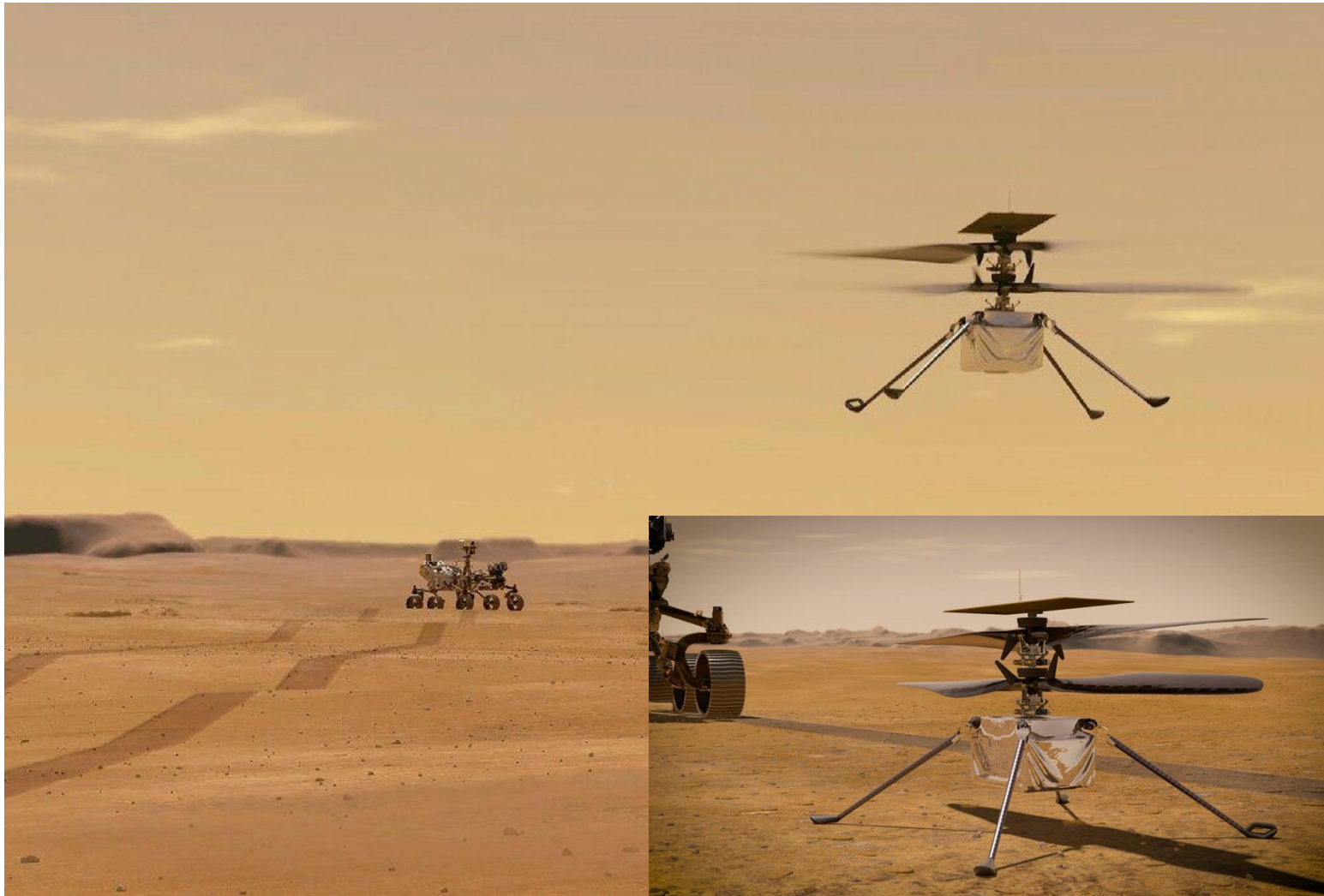


[NASA'S Mars Exploration Rover Spirit](#) captured this westward view from atop a low plateau where Spirit spent the closing months of 2007.

Vision systems (JPL) used for several tasks

- Panorama stitching
- 3D terrain modeling
- Obstacle detection, position tracking
- For more, read “Computer Vision on Mars” by Matthies et al.

Mars Exploration



Feb. 18, 2021, Jezero Crater, Mars

Perseverance rover and Ingenuity Helicopter on Mars

Techniques behind Mars Helicopter

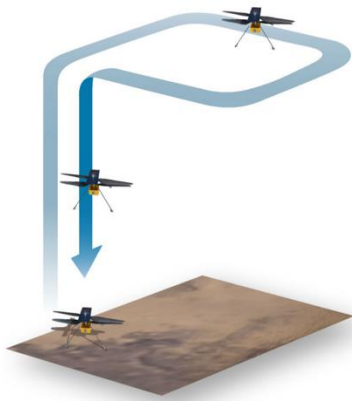
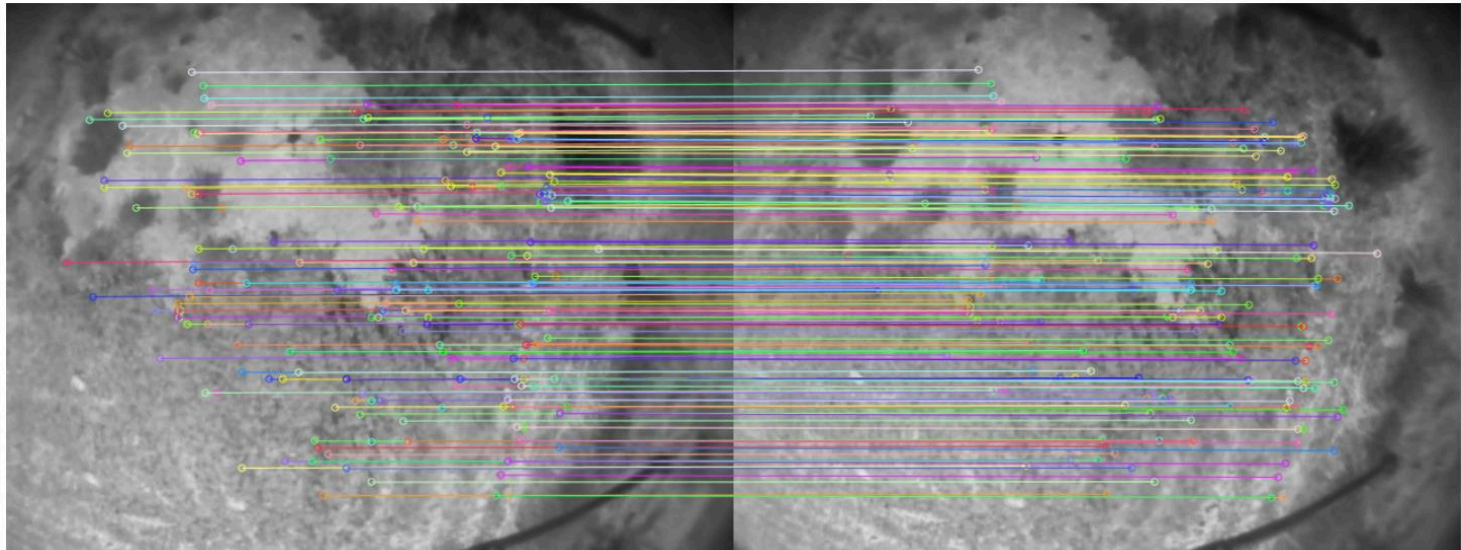
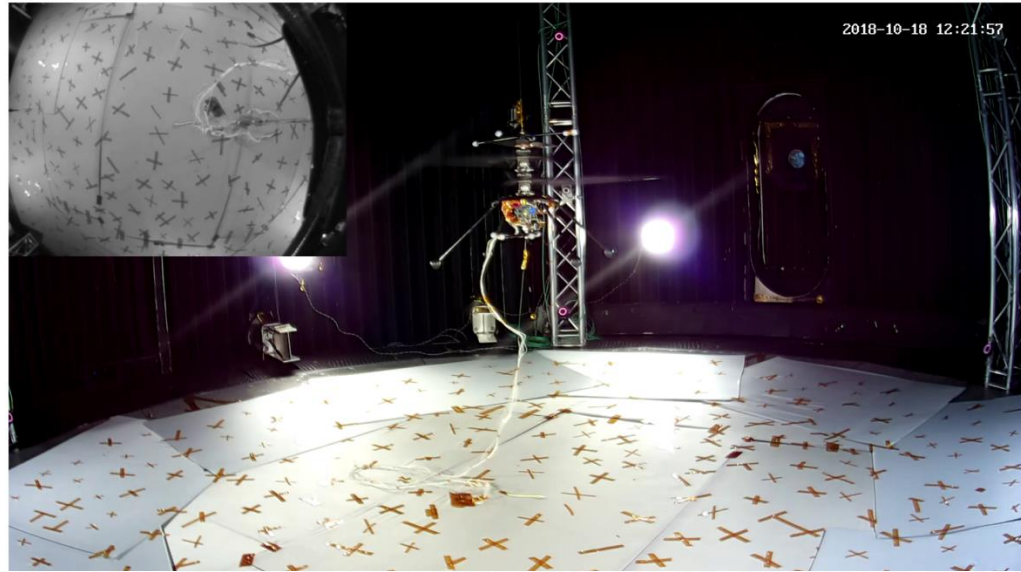


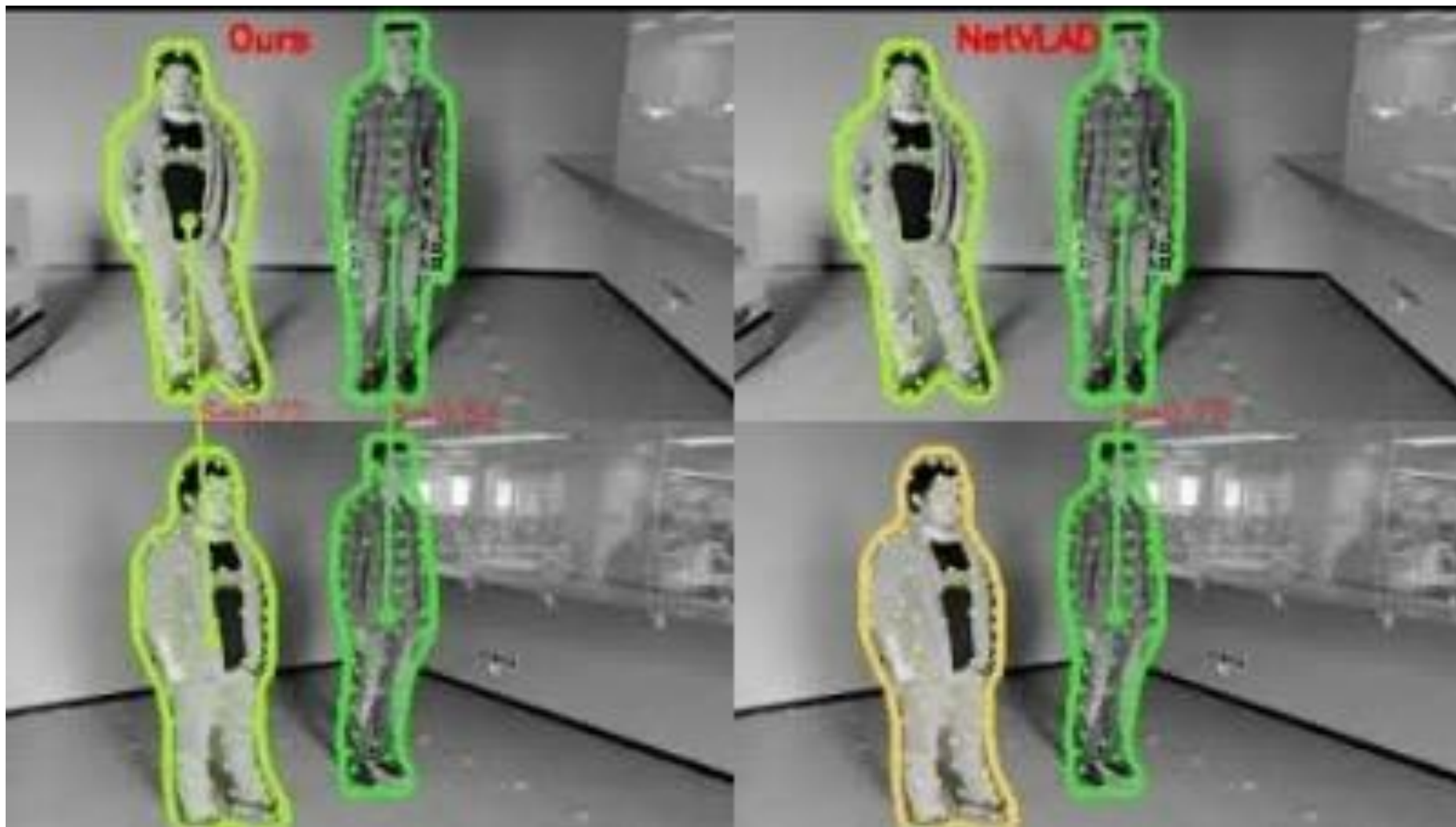
Fig. 2 Illustration of a Mars Helicopter flight, beginning and ending in the same pre-inspected safe area



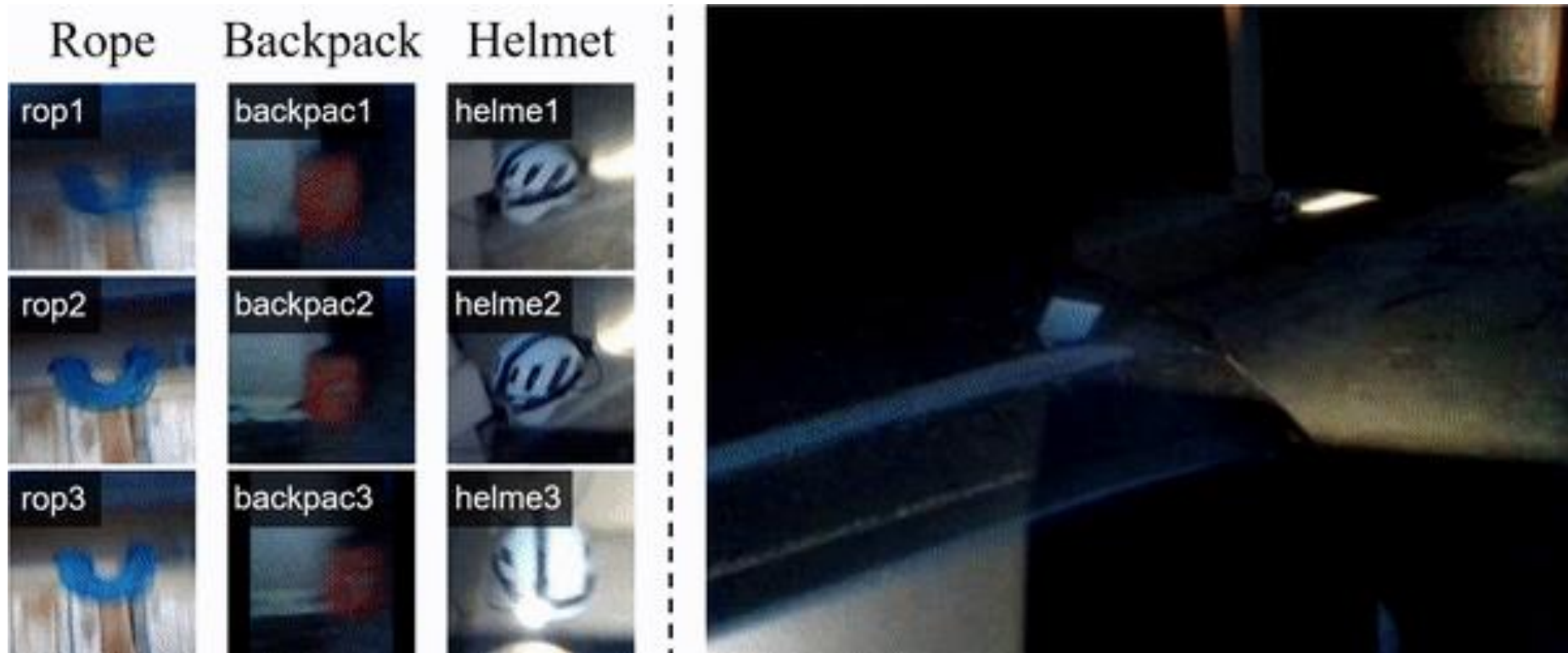
Top Venues You should know..

- Computer Vision
 - IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)
 - International Conf. on Computer Vision (ICCV)
 - European Conf. on Computer Vision (ECCV)
- Robotics (Check <https://roboranking.org>)
 - Science Robotics
 - Transactions on Robotics (T-RO)
 - International Journal of Robotics Research (IJRR)
 - Robotics: Science and Systems (RSS)
 - Robotics and Automation Letters (RA-L)
 - International Conf. on Robotics and Automation (ICRA)
 - International Conf. on Intelligent Robots and Systems (IROS)
- Machine Learning
 - Neural Information Processing Systems (NeurIPS)
 - International Conference on Machine Learning (ICML)
 - International Conference on Learning Representations (ICLR)

Person Re-identification

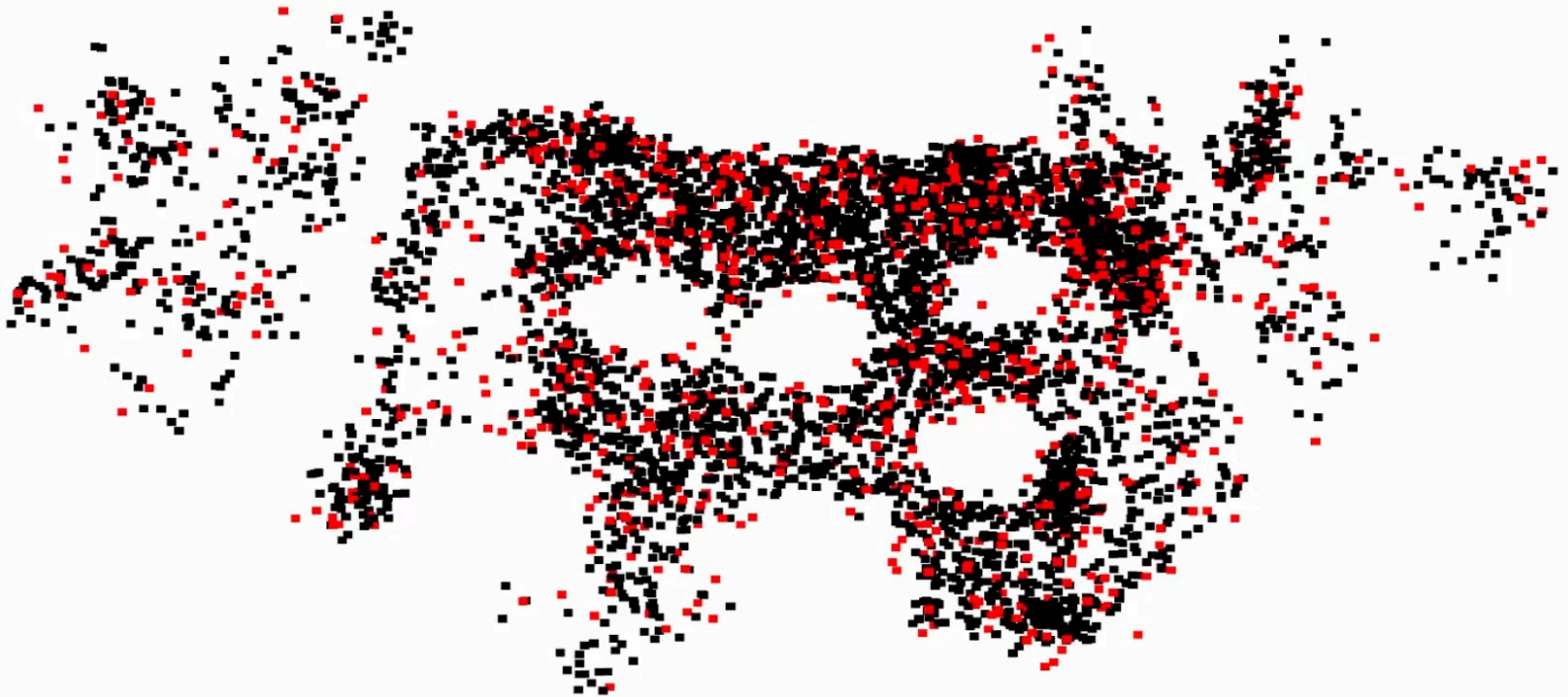


Search and Rescue (Few-shot Detection)



Point Clouds

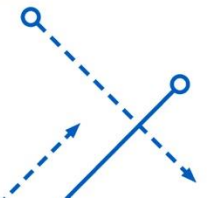
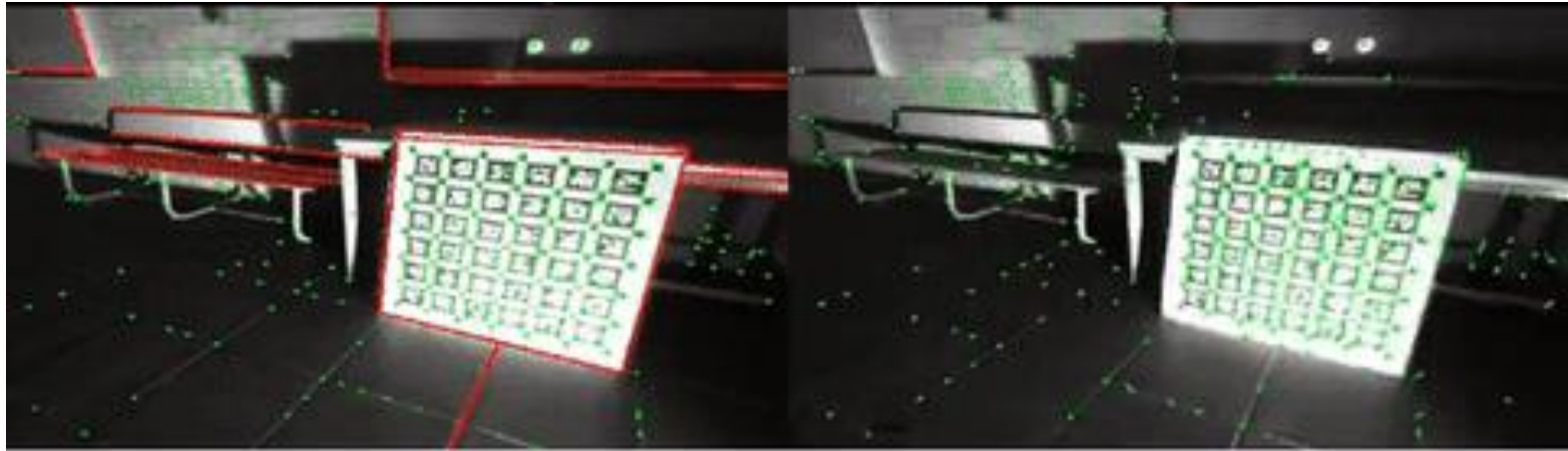
Completion + Denoising + Upsampling + Colorization



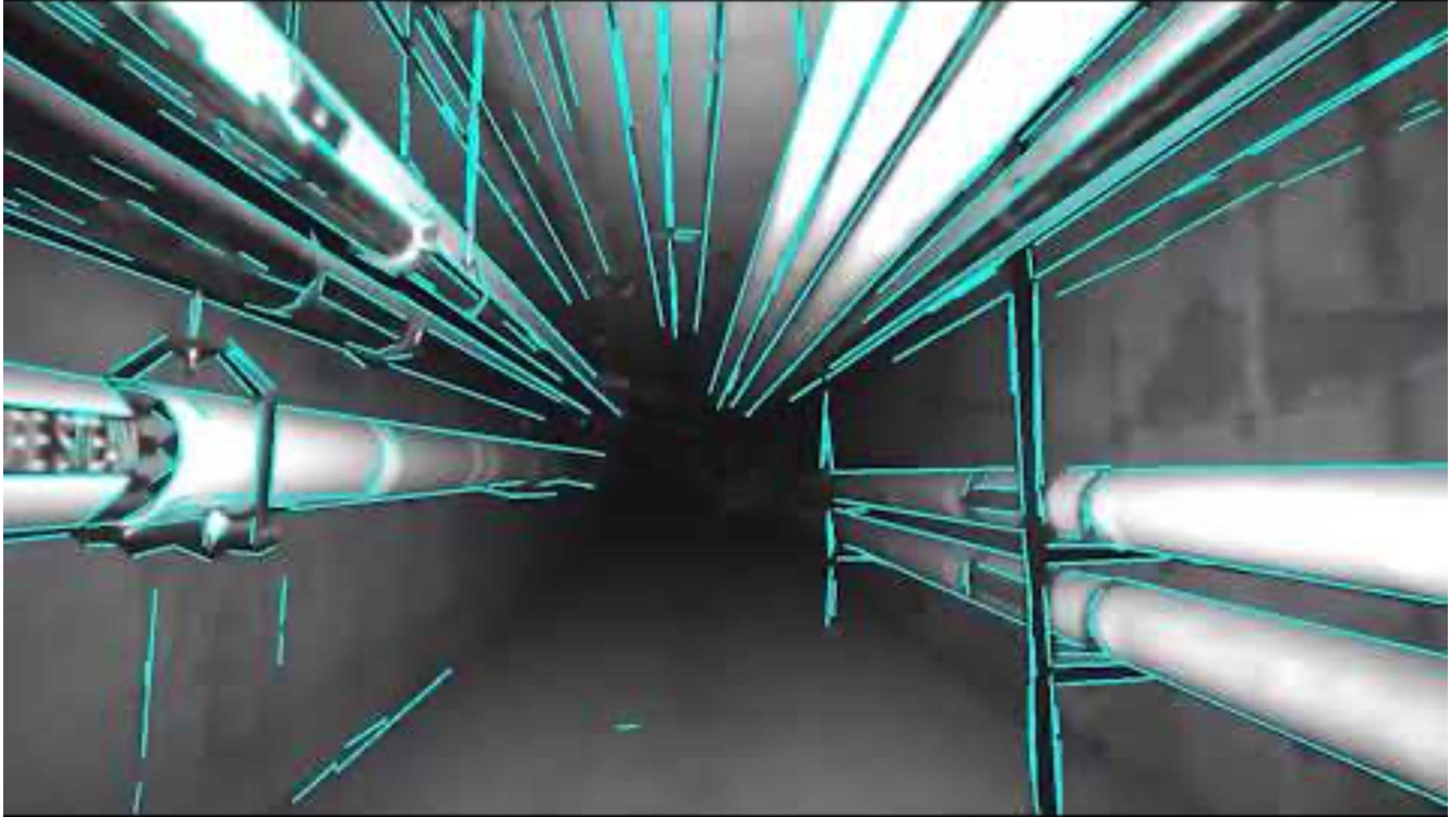
Simultaneously Localization and Mapping (SLAM)



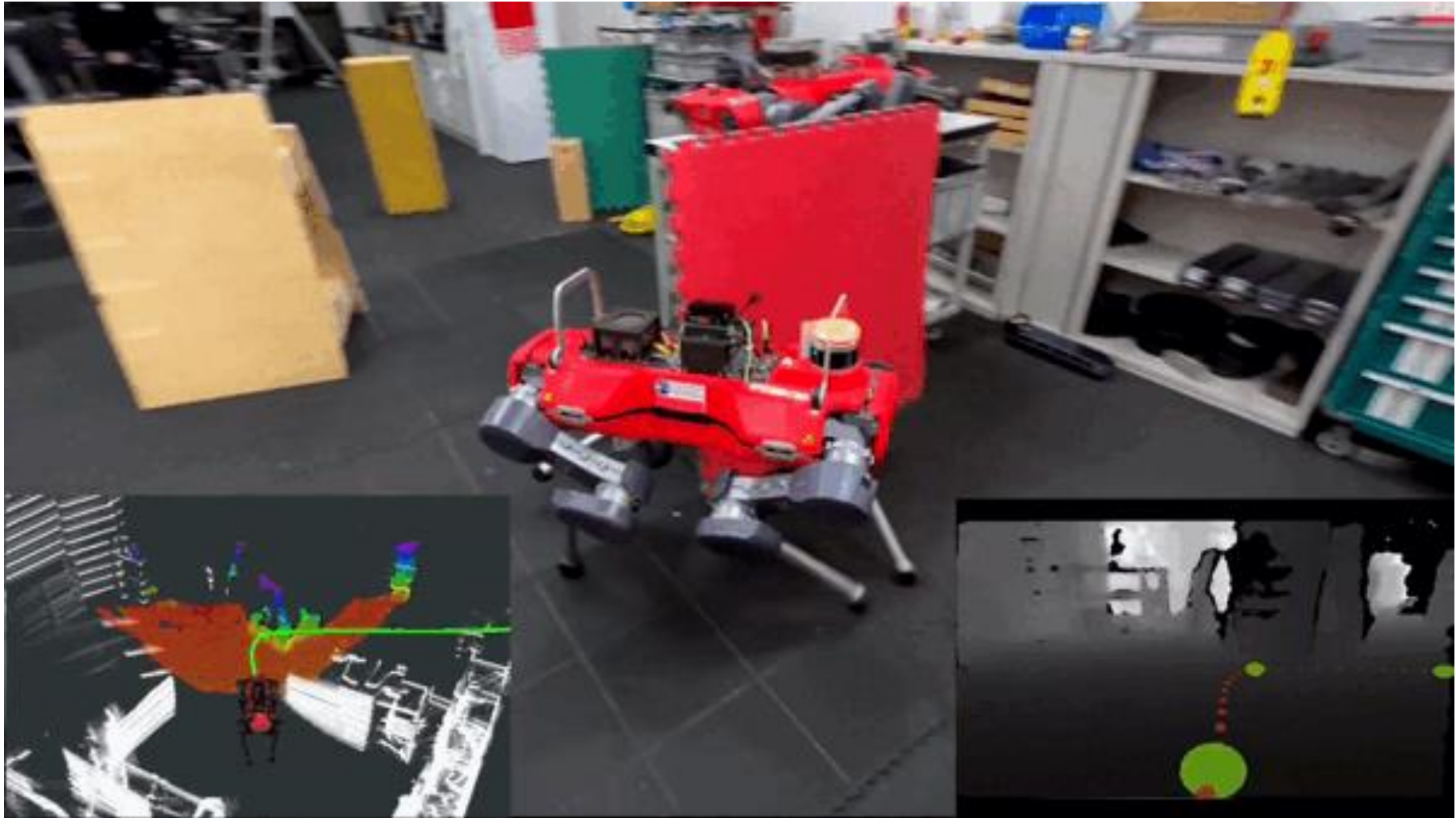
Visual Odometry



Line Detection



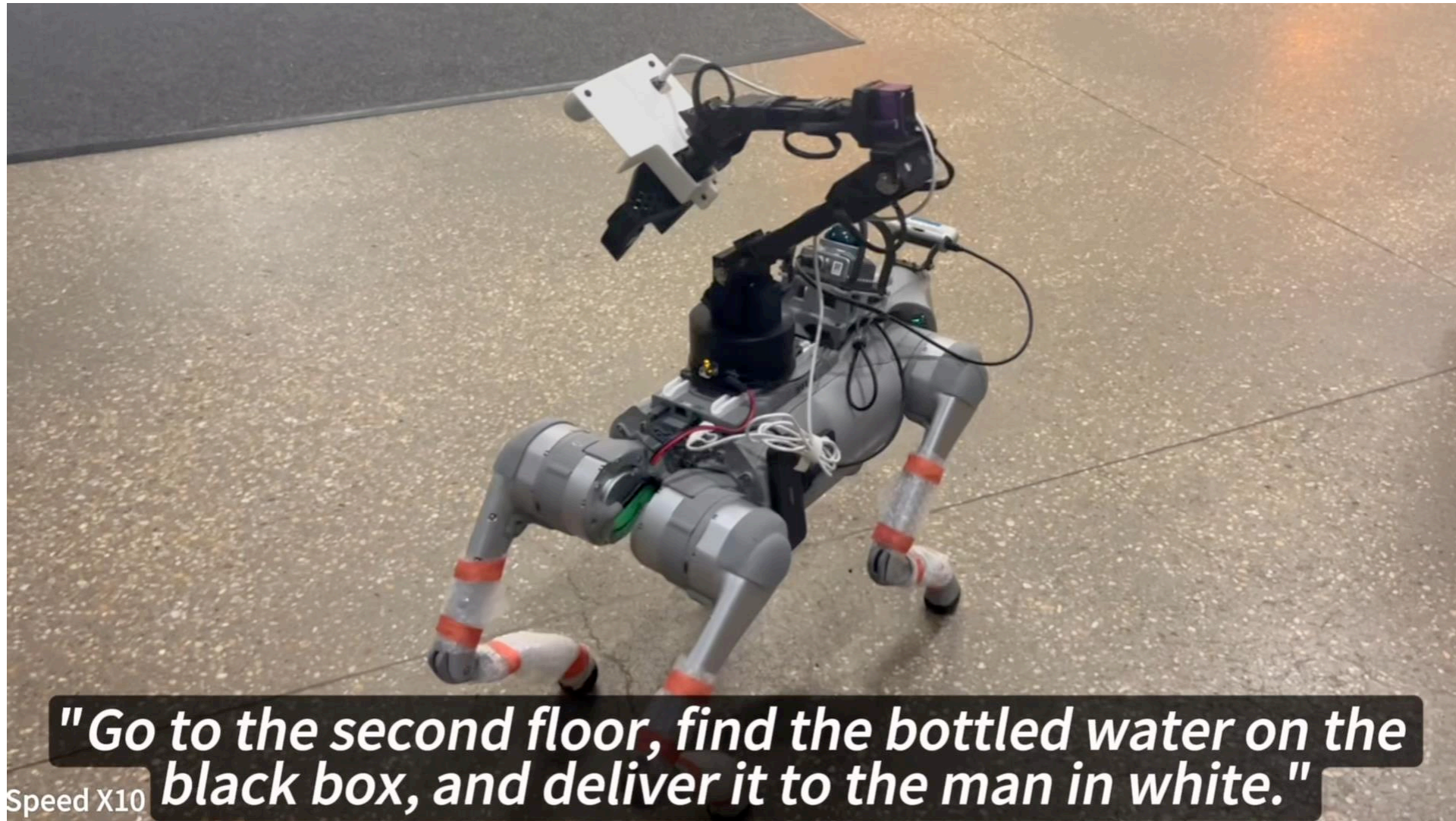
Path Planning



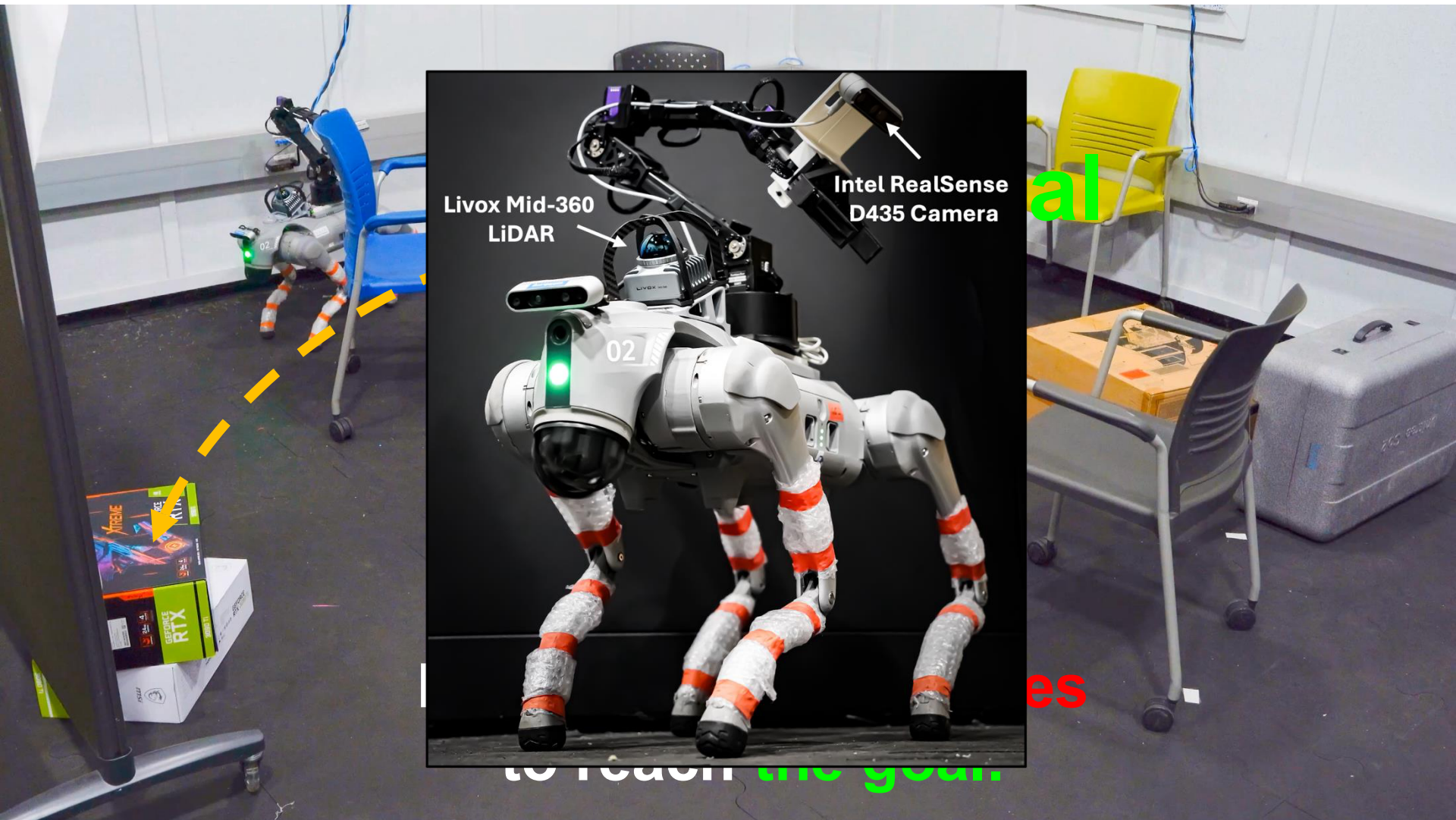
Vision-language Navigation



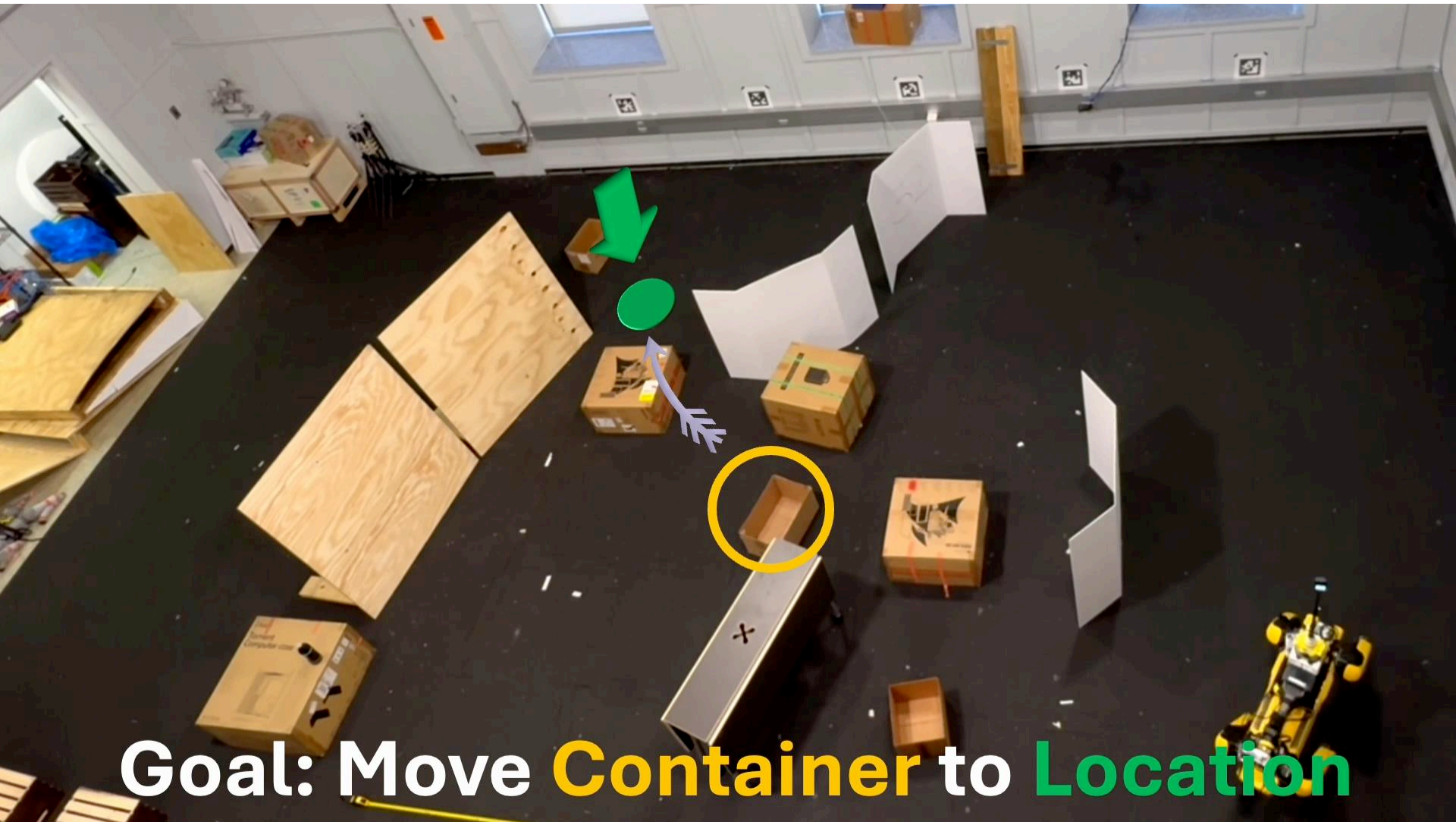
Vision-language Navigation



Fast Task Planning



Long-horizon Task Planning



Goal: Move **Container** to **Location**

Last but not least

- Quiz 0
 - Help you familiar with UB learn system.
 - We'll take a picture for memories!